Quantum State Discrimination with Bosonic Channels and Gaussian States

Course: PH3203 Advanced Quantum Mechanics
Instructor: Professor Sourin Das

A Term Paper Review of Chapters 1–3[1]

Based on the MIT Doctoral Thesis by Si Hui Tan (2010)

Authors

Sapnodip Pramanik
23MS145

Bikash Jyoti Saharia
23MS133

Bhargab Neog
23MS177

Shounak Goswami
23MS158

April 27, 2026

INDIAN INSTITUTE OF SCIENCE EDUCATION AND RESEARCH, KOLKATA

Acknowledgements

[ACKNOWLEDGEMENT]

Abstract

[ABSTRACT]

Contents

1 Introduction **4**

1.1 The notion of a quantum state . . . 4

1.2 Measuring a quantum state . . . 4

1.3 State Discrimination . . . 7

1.3.1 State Discrimination with Minimum Error . . . 8

1.3.2 Unambiguous State Discrimination . . . 8

1.3.3 Pretty-Good Measurement . . . 10

1.3.4 Other Measurement Strategies . . . 11

1.4 The Role of State Discrimination in Quantum Information Theory . . . 11

1.5 Continuous Variables in Quantum Mechanics . . . 13

1.6 Scope of This Work . . . 14

2 Continuous Variable Quantum Information Theory **15**

2.1 Introduction . . . 15

2.2 Quantization of a Single Electromagnetic Field Mode . . . 16

2.2.1 Hamiltonian of the Field . . . 17

2.3 Bosonic Modes from the Harmonic Oscillator . . . 18

2.3.1 Connection to the Standard Harmonic Oscillator . . . 18

2.3.2 Creation and Annihilation Operators . . . 19

2.3.3 Hamiltonian in Terms of Ladder Operators . . . 19

2.3.4 Fock States and Vacuum State . . . 20

2.3.5 Multi-Mode Bosonic System . . . 20

2.3.6 Quadrature Operators for Each Mode . . . 21

2.3.7 Phase Space Representation . . . 21

2.4 Mean Vector, Expectation Values and the Covariance Matrix . . . 22

2.4.1 Covariance Matrix . . . 22

2.5 The Symplectic Structure of Phase Space and the Group $\mathrm{Sp}(2n, \mathbb{R})$ . . . 23

2.5.1 Definition of Symplectic Transformations . . . 24

2.5.2 The Symplectic Group . . . 24

2.5.3 Real Canonical Linear Transformations . . . 24

2.6 Quadratic Hamiltonians and Gaussian Transformations . . . 25

2.7 Characteristic Functions (Quantum) . . . 28

2.8 Wigner Function . . . 29

2.9 P Representation . . . 30

2.10 Trace Rules in Phase Space . . . 30

2.11 Uncertainty Principle . . . 31

2.12 Coherent States . . . 32

2.12.1 Fock representation of the coherent state 32
2.12.2 The coherent state as a displaced vacuum state - the displacement operator 34
2.13 Squeeze Operator 36
2.13.1 The unitary squeeze operator 36
2.13.2 Defining the Operators 37
2.13.3 Applying the Baker-Campbell-Hausdorff Lemma 37
2.13.4 Calculating the Nested Commutators 37
2.13.5 Summing the BCH Series 38
2.13.6 Final Expression 39
2.14 Derivation of the Normal-Ordered Squeezing Operator using Operator Algebra 39
2.14.1 Constructing the Normal-Ordered Ansatz 39
2.14.2 Pushing the Creation Operator Through the Ansatz 40
2.14.3 Pushing the Annihilation Operator Through the Ansatz 41
2.14.4 Determining the Normalization Scalar 41
2.14.5 Final Assembly 42
2.14.6 Squeezed state form 42
2.15 Derivation of the Wigner Function for the Squeezed Vacuum State 44
2.15.1 The Annihilation Condition 44
2.15.2 Formulating $\hat{b}$ in Phase Space 44
2.15.3 Solving the Position Wavefunction 45
2.15.4 Integrating the Wigner Function 45
2.16 Algebraic Derivation of Two-Mode Squeezing 47
2.16.1 Definitions 47
2.16.2 Mode Evolution 47
2.16.3 Normal-Ordered Ladder Expansion 48
2.16.4 Sidenote : 49
2.17 Creating Gaussian EPR states 50
2.17.1 Preliminary Properties for Usage 50
2.17.2 The First Moments (Means) 50
2.17.3 The Photon Number 50
2.17.4 The Variance / Squeezing Terms 51
2.18 Explicit Derivation of the Matrices for the Gaussian EPR State 52
2.18.1 Deriving the Input Covariance Matrix 52
2.18.2 Deriving the Symplectic Beam-Splitter Matrix 53
2.18.3 Deriving the Output EPR Covariance Matrix 53

3 State Discrimination Using Gaussian States **54**
3.1 Introduction 54
3.2 Normal form decomposition 56
3.3 Distinguisability Of States 61
3.3.1 Helstrom's Optimal Discrimination 61
3.3.2 Fidelity 63
3.3.3 Quantum Chernoff Bounds 64
3.4 Fidelity for Gaussian States 66
3.4.1 Single Mode Gaussian States 66
3.4.2 Multimode Gaussian States 67
3.5 Quantum Chernoff Bounds for Gaussian States 67
3.5.1 Numerical Analysis of Quantum Chernoff Bounds 69
3.5.2 Results 72

References **72**

1 Introduction

1.1 The notion of a quantum state

A quantum state represented by a positive Hermitian operator $\hat{\rho}$ with unit trace, contains all physically relevant information about a quantum system. If two quantum systems are prepared in identical states, any measurement performed on them will yield identical outcome probabilities, making the different preparation methods observationally equivalent. Conversely, if the systems are in different quantum states, there necessarily exists at least one measurement that produces different probability distributions, allowing their preparations to be distinguished. The systematic process of designing and performing measurements to identify which of several possible states (and thus which preparation) a given system is in is called quantum state discrimination.

1.2 Measuring a quantum state

While the quantum state itself is an abstract mathematical object, measurements provide it with concrete operational meaning. The outcomes of quantum measurements are always classical, such as a photodetector click count or an ion's position in a trap, and these outcomes behave as random variables. By performing the same measurement many times on identically prepared quantum systems, one can reconstruct the statistical distribution of outcomes. If this process is repeated for every possible measurement, the complete set of outcome probability distributions fully characterizes the quantum state. Thus, the state is not directly observable, but its empirical reality is revealed through the statistics of measurement outcomes, bridging the gap between abstract formalism and experimental practice. The formal relationship between quantum states and measurements is captured by the postulates of standard quantum measurement theory. These postulates can be summarized as follows:

1. Every physical observable corresponds to a Hermitian operator $\hat{X}$ with a spectral decomposition
$$\hat{X} = \sum_j \lambda_j |j\rangle\langle j| \tag{1}$$
where λ_j are the possible measurement outcomes.

2. The projectors
$$P_j = |j\rangle\langle j| \tag{2}$$
associated with the eigenstates form a complete set, satisfying
$$\sum_j P_j = \mathbb{I} \tag{3}$$

and are orthogonal such that

$$P_i P_j = \delta_{ij} P_i \tag{4}$$

3. A measurement of $\hat{X}$ yields one of its eigenvalues λ_j as the result.

4. If the outcome λ_j is obtained, the system collapses to the post-measurement state

$$\hat{\rho}_j = \frac{P_j \hat{\rho} P_j}{\mathrm{tr}(P_j \hat{\rho} P_j)} \tag{5}$$

5. The probability of obtaining the outcome λ_j is given by

$$p_j = \mathrm{tr}(\hat{\rho} P_j) \tag{6}$$

6. When the measurement is performed but the outcome is not recorded, the post-measurement state is described by the density operator

$$\hat{\rho}' = \sum_j P_j \hat{\rho} P_j \tag{7}$$

representing an average over all possible outcomes.

It is important to note that in stating the postulates, the Hilbert space has been assumed to be finite and discrete solely for the sake of notational simplicity. In principle, these postulates apply equally to states with infinite-dimensional Hilbert spaces, which is precisely the case for continuous variable systems. The postulates make it clear that the measurement process is fundamentally random, with the outcome able to take a range of values. The number of possible measurement outcomes is equal to the dimension of the Hilbert space of the state. The projection operators P_j used in this formulation are referred to as projective measurements or von Neumann measurements. However, it turns out that the requirement that measurement operators be strictly projective can be relaxed while still maintaining a meaningful statistical distribution for the outcomes. In particular, Postulate 3 provides an algorithm for generating outcome probabilities.

If we replace the projective measurement operator P_j with a positive operator Π_j that may not necessarily be a projection, the probability of obtaining outcome j is given by

$$p_j = \mathrm{tr}(\hat{\rho} \Pi_j) \tag{8}$$

Since Π_j is a positive operator, $p_j \geq 0$, and if we further require that the set of such operators for a finite set of outcomes is complete, meaning $\sum_j \Pi_j = \mathbb{I}$, then the quantities p_j can be interpreted as probabilities for the outcomes of the measurement described by

the operator set $\{\Pi_j\}$. Such a set of measurement operators is called a positive operator valued measure, or POVM. A key feature of POVMs is that their elements do not need to be orthogonal. Consequently, the number of possible outcomes can be larger than the dimension of the Hilbert space of the state being measured, offering greater flexibility in measurement design.

Using POVMs, we can replace the postulates of the quantum theory of standard measurements with the following more flexible ones:

1. Measurement operators are positive operators $\Pi_j \geq 0$ where the set of all such operators is complete: $\sum_j \Pi_j = \mathbb{I}$. The set of such operators is called a POVM. The elements of the POVM can be written in terms of generalized measurement operators A_j, where
$$\Pi_j = A_j^\dagger A_j \tag{9}$$
The generalized measurement operators can be non-Hermitian, though the elements of the POVM must be Hermitian by construction.

2. Each element of the POVM set represents a possible outcome of the measurement.

3. The state of the system after the measurement is
$$\hat{\rho}_j = \frac{A_j \hat{\rho} A_j^\dagger}{\mathrm{tr}(A_j \hat{\rho} A_j^\dagger)} \tag{10}$$

4. The probability that the outcome j (indexing the measurement operator Π_j) is found is
$$p_j = \mathrm{tr}(\Pi_j \hat{\rho}) \tag{11}$$

5. If we perform the measurement but do not record the results, the post-measurement state is described by the density operator
$$\hat{\rho}' = \sum_j A_j \hat{\rho} A_j^\dagger \tag{12}$$

The POVM formalism described above provides a way to describe classical information obtained from a quantum state by way of a measurement. State discrimination is the process of processing this information to determine, to the best of our knowledge, which quantum state we possess. The fundamental question of how well we can discriminate between quantum states lies at the heart of quantum mechanics. When the set of possible states are orthogonal to one another, perfect discrimination is always possible. This can be achieved by choosing a set of projection operators that correspond exactly to the

set of possible states. However, when the possible states are non-orthogonal, perfect discrimination is impossible. In such cases, we must define optimal discrimination with respect to some figure of merit, though finding this optimal discrimination is often non-trivial. The chapter will next set up the problem of state discrimination and describe two commonly used strategies: minimum-error discrimination and unambiguous discrimination. Following that, some interesting well-known measurements that do not fall into these two categories, such as the pretty-good measurement and the no-measurement strategy, will be discussed.

1.3 State Discrimination

Consider an ensemble of quantum systems such that each system is prepared in one of N possible states, say $\hat{\rho}_1, \ldots, \hat{\rho}_N$, with a priori probabilities $p_1, \ldots, p_N$ respectively, where $\sum_{j=1}^{N} p_j = 1$. In general, the states can be pure or mixed, and they can be either discrete variable states or continuous variable states. The observer performing state discrimination has full knowledge of the ensemble being prepared, meaning he knows the possible states and their prior probabilities. However, he does not know which one of the N states he possesses, and he must make a measurement to determine this. State discrimination describes how he should design and perform these measurements to optimally identify the unknown state.

There are two commonly used strategies for state discrimination. The first is discrimination with minimum error, attributed to Helstrom, who studied distinguishing between two given quantum states. Helstrom found an optimum measurement strategy for both mixed and pure states that minimizes the probability of error. This optimum measurement strategy is known as the Helstrom measurement, and the resulting minimum error is called the Helstrom bound. The general problem of discriminating between $N > 2$ states with minimum error is considerably more difficult. Although the necessary and sufficient conditions for an optimal measurement have been found, how to implement such an optimal measurement remains an open problem, except for a few special cases.

The second strategy is unambiguous discrimination, first proposed by Ivanovic. This strategy allows perfect discrimination between two non-orthogonal states, but with the caveat that a non-conclusive (inconclusive) answer is permitted. The specific case of discriminating between two pure non-orthogonal states was solved by Dieks and Peres. However, the general case of unambiguously discriminating between N pure states remains an open problem. It has been shown by Chefles that unambiguous discrimination is only possible if the states to be distinguished are linearly independent. Additionally, there are active research areas focused on extending unambiguous discrimination strategies to

mixed states.

Having described the two strategies briefly, let us now discuss each of them in more detail.

1.3.1 State Discrimination with Minimum Error

In the minimum error discrimination strategy, we always reach a conclusion about which state was prepared, though errors in this conclusion are inevitable. The goal of the optimal strategy is to minimize the detection error. To formulate this as a definite optimization problem, we must define what constitutes a detection error.

Let Π_j be the POVM element corresponding to the state $\hat{\rho}_j$ being prepared. According to the postulates of quantum measurement theory, if the actual prepared state is $\hat{\rho}$, the probability of guessing that the state is $\hat{\rho}_j$ is $\mathrm{tr}(\hat{\rho}\Pi_j)$. Since the state $\hat{\rho}_j$ occurs with a priori probability p_j, the overall probability of a correct guess is given by

$$P_{\text{corr}} = \sum_{j=1}^{N} p_j \,\mathrm{tr}(\hat{\rho}_j \Pi_j) \tag{13}$$

The detection error is then defined as

$$P_{\text{err}} = 1 - P_{\text{corr}} \tag{14}$$

To minimize the error, we must minimize P_{err} over all possible sets of POVM elements $\{\Pi_j\}$ for $j = 1, \ldots, N$ subject to the constraints $\Pi_j \geq 0$ and $\sum_{j=1}^{N} \Pi_j = \mathbb{I}$. The POVM that achieves this minimization is called the optimal measurement, and the resulting error is the minimum error. The special case of distinguishing single copies of two mixed states will be derived later in Section 3.3.1.

1.3.2 Unambiguous State Discrimination

Unlike discrimination with minimum error, in unambiguous discrimination we are not permitted to make measurement errors. However, this strategy does not succeed 100% of the time, and this trade-off is necessary to achieve error-free discrimination.

To illustrate how unambiguous state discrimination works, we consider a simple example involving two pure states $|\psi_0\rangle$ and $|\psi_1\rangle$ occurring with equal a priori probabilities, given by

$$|\psi_0\rangle = \cos\theta|0\rangle + \sin\theta|1\rangle \tag{15}$$

$$|\psi_1\rangle = \cos\theta|0\rangle - \sin\theta|1\rangle \tag{16}$$

These states are non-orthogonal with overlap $\langle\psi_0|\psi_1\rangle = \cos^2\theta - \sin^2\theta = \cos(2\theta)$.

A POVM for distinguishing between these states can be constructed with elements

$$\Pi_0 = a_0(\sin\theta|0\rangle + \cos\theta|1\rangle)(\sin\theta|0\rangle + \cos\theta|1\rangle) \tag{17}$$

$$\Pi_1 = a_1(\sin\theta|0\rangle - \cos\theta|1\rangle)(\sin\theta|0\rangle - \cos\theta|1\rangle) \tag{18}$$

$$\Pi_2 = \mathbb{I} - \Pi_0 - \Pi_1 \tag{19}$$

where the coefficients are chosen such that

$$\langle\psi_0|\Pi_1|\psi_0\rangle = \langle\psi_1|\Pi_0|\psi_1\rangle = 0 \tag{20}$$

and $0 \leq a_0, a_1 \leq 1$. When outcome 0 is obtained, we can confidently conclude the state is $|\psi_0\rangle$; when outcome 1 is obtained, we can confidently conclude the state is $|\psi_1\rangle$. However, if outcome 2 occurs, no definite conclusion can be made.

To evaluate how effective this unambiguous discrimination strategy is, we need to determine the probability that the measurement yields an inconclusive result. A lower probability of inconclusive outcomes indicates a better discrimination scheme. Therefore, we now derive this probability and then optimize it over the available parameters. We begin by computing the conclusive probabilities. we first compute the conclusive probabilities. The probability of obtaining outcome 0 when $|\psi_0\rangle$ is sent is

$$\langle\psi_0|\Pi_0|\psi_0\rangle = a_0|\langle\psi_0|(\sin\theta|0\rangle + \cos\theta|1\rangle)|^2 \tag{21}$$

The overlap is

$$\langle\psi_0|(\sin\theta|0\rangle + \cos\theta|1\rangle) = \cos\theta\sin\theta + \sin\theta\cos\theta = \sin(2\theta) \tag{22}$$

Thus,

$$\langle\psi_0|\Pi_0|\psi_0\rangle = a_0\sin^2(2\theta) \tag{23}$$

Similarly, for $|\psi_1\rangle$,

$$\langle\psi_1|\Pi_1|\psi_1\rangle = a_1\sin^2(2\theta) \tag{24}$$

Since the prior probabilities are equal ($p_0 = p_1 = 1/2$), the overall probability of a conclusive result is

$$P_{\text{conc}} = \frac{1}{2}a_0 \sin^2(2\theta) + \frac{1}{2}a_1 \sin^2(2\theta) \tag{25}$$

Completeness of the POVM and positivity of Π_2 imposes constraints on a_0 and a_1. The optimal choice that maximizes the conclusive probability (or equivalently minimizes the inconclusive probability) is found to be $a_0 = a_1 = 1/(1 + \cos(2\theta))$. Substituting this value, we obtain

$$\langle\psi_0|\Pi_0|\psi_0\rangle = \frac{\sin^2(2\theta)}{1 + \cos(2\theta)} = \frac{1 - \cos^2(2\theta)}{1 + \cos(2\theta)} = 1 - \cos(2\theta) \tag{26}$$

Thus, the probability of obtaining an inconclusive result when $|\psi_0\rangle$ is sent is $1 - (1 - \cos(2\theta)) = \cos(2\theta)$. The same holds for $|\psi_1\rangle$. Therefore, the overall probability of an inconclusive result is

$$P(\text{inconclusive result}) = \cos(2\theta) \tag{27}$$

This minimum value is known as the Ivanovic-Dieks-Peres (IDP) limit. Analytical solutions for the IDP limit are not known for the general case of $N > 2$ pure states, though the solution for three pure states has been derived. Various numerical techniques have also been developed to calculate the IDP limit in the general case.

1.3.3 Pretty-Good Measurement

More often than not, in minimum-error state discrimination, it is not possible to find the optimal measurement. In such cases, the pretty-good measurement (PGM), also known as the square-root measurement, is—as its name suggests—a good measurement strategy to pursue. For a given set of N possible states, we can always construct the POVM elements of the PGM as

$$\hat{\Pi}_i = p_i \hat{\rho}^{-1/2} \hat{\rho}_i \hat{\rho}^{-1/2} \text{ for } i = 1, \ldots, N \tag{28}$$

where

$$\hat{\rho} = \sum_i p_i \hat{\rho}_i \tag{29}$$

and p_i is the probability of the ith state $\hat{\rho}_i$ occurring. The POVM elements defined in this way satisfy both positivity and completeness. Although the PGM is not the optimal measurement in the general case, it turns out to be optimal for many of the cases in which the minimum-error measurement is known. These results have been summarized in the literature.

1.3.4 Other Measurement Strategies

An interesting state discrimination strategy is the no-measurement strategy. For some sets of states, the optimal measurement is to make no measurements at all and simply guess the state that is most likely. A practical example where the no-measurement strategy applies is distinguishing a state from random noise: if the signal-to-noise ratio is sufficiently low, one should always guess that the received state was random noise.

Additionally, there exist mixed strategies that allow inconclusive results while minimizing the probability of detection error, as considered by Chefles and Barnett. The suggested potential use of such mixed strategies is to provide added flexibility for an eavesdropper to "listen in" on a cryptographic protocol.

Having discussed some of the strategies for state discrimination, important questions still remain: How does state discrimination help our understanding of quantum mechanics, and how can it be used in quantum protocols? The next section will address these questions.

1.4 The Role of State Discrimination in Quantum Information Theory

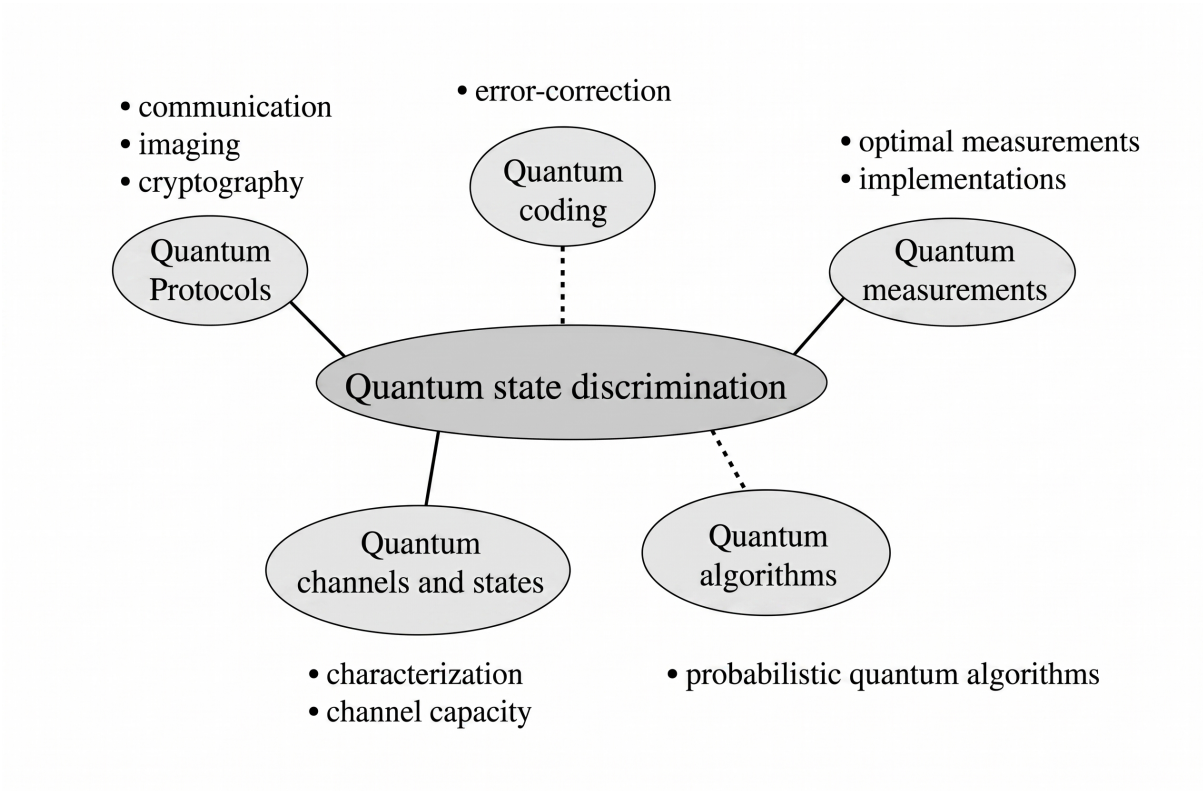

Figure 1: An overview of the central role of quantum state discrimination in quantum information theory.

Quantum state discrimination lies at the heart of quantum computation and communication. The specific strategy of state discrimination applied depends on the task in quantum information theory to be solved. For example, using state discrimination techniques, we can show that quantum mechanics does not permit perfect discrimination between nonorthogonal states $|\psi_0\rangle$ and $|\psi_1\rangle$—a result that is also directly implied by the no-cloning theorem.

To demonstrate this, assume we have a POVM with two elements Π_0 and Π_1. For perfect discrimination, we require that

$$\Pi_0|\psi_1\rangle = 0 \tag{30}$$

$$\Pi_1|\psi_0\rangle = 0 \tag{31}$$

This means that if outcome 0 is obtained, we deduce the state was $|\psi_0\rangle$, and if outcome 1 is obtained, we deduce the state was $|\psi_1\rangle$. The POVM operators must also satisfy the completeness relation

$$\Pi_0 + \Pi_1 = \mathbb{I} \tag{32}$$

To derive the necessary condition for perfect discrimination, we multiply the above equation on the left by $\langle\psi_0|$ and on the right by $|\psi_1\rangle$:

$$\langle\psi_0|(\Pi_0 + \Pi_1)|\psi_1\rangle = \langle\psi_0|\mathbb{I}|\psi_1\rangle \tag{33}$$

Distributing the left-hand side gives

$$\langle\psi_0|\Pi_0|\psi_1\rangle + \langle\psi_0|\Pi_1|\psi_1\rangle = \langle\psi_0|\psi_1\rangle \tag{34}$$

Now, from the perfect discrimination conditions, we have $\Pi_0|\psi_1\rangle = 0$. Therefore, the first term $\langle\psi_0|\Pi_0|\psi_1\rangle = 0$. For the second term, note that Π_1 is a positive operator. If $\Pi_1|\psi_0\rangle = 0$, then taking the adjoint gives $\langle\psi_0|\Pi_1 = 0$. However, this does not directly imply $\langle\psi_0|\Pi_1|\psi_1\rangle = 0$. In standard unambiguous discrimination, the POVM elements are constructed such that Π_1 projects onto a subspace orthogonal to $|\psi_0\rangle$, which indeed forces $\langle\psi_0|\Pi_1|\psi_1\rangle = 0$. Under this construction, we have,

$$0 + 0 = \langle\psi_0|\psi_1\rangle \tag{35}$$

Thus,

$$\langle\psi_0|\psi_1\rangle = 0 \tag{1.20}$$

This condition can only be satisfied if the states $|\psi_0\rangle$ and $|\psi_1\rangle$ are orthogonal. Conse-

quently, perfect discrimination is impossible for non-orthogonal states. Insights such as these are prevalent when applying state discrimination to problems in quantum information theory. The fact that we cannot perfectly distinguish between non-orthogonal states is what makes quantum key distribution possible.

Another important use of state discrimination is to characterize properties of quantum states, such as entanglement and separability. Much research has been devoted to this area. One of the main motivations behind such studies is to determine whether entangled states are useful for certain tasks, or whether entanglement is needed for a given task at all. Quantum metrology, quantum imaging, and the classical capacity of quantum channels are examples of such investigations.

State discrimination is also indispensable in quantum error correction. Error correction cannot be performed without syndrome measurement. A syndrome measurement determines whether a quantum state has been corrupted and, if so, which part of the state has been affected. The syndrome measurement is typically a multi-state measurement that does not disturb the encoded state but still provides information about the error. Such measurements lie at the heart of error correction.

Finally, state discrimination has found uses in quantum algorithms through a variant known as quantum state filtering. In quantum state filtering, the goal is to determine whether an unknown quantum state is a specific state $|\psi_k\rangle$ from a set of known states $\{|\psi_1\rangle, \ldots, |\psi_N\rangle\}$, or one of the others. Using optimal solutions in quantum state filtering, it is possible to implement an efficient probabilistic quantum algorithm that distinguishes between sets of Boolean functions, which generalizes the Deutsch-Jozsa algorithm. It has also been proposed that quantum decision problems can be characterized by state discrimination.

Having discussed the role of state discrimination in the major topics of quantum information theory, we now introduce the encoding of information onto continuous variable states.

1.5 Continuous Variables in Quantum Mechanics

While quantum information is primarily concerned with discrete quantum variables, such as qubits, continuous variable states have always played an important role in quantum mechanics. The position and momentum of the quantized harmonic oscillator are continuous variables. Because a mode of the electromagnetic field can be thought of as a harmonic oscillator, its quadrature amplitudes (the electric and magnetic fields) are continuous

variables. Due to this connection to the quantized electromagnetic field, the development of the laser in the 1960s inspired considerable research into the properties of continuous quantum variables.

Continuous variables also play an important role in quantum information. The original EPR thought experiment was formulated in terms of the entangled momentum and position states of two particles. Furthermore, one of the first experimental demonstrations of quantum teleportation was performed using continuous variable states. Many aspects of discrete quantum information theory have continuous analogues, including quantum codes. It is even possible to formulate a theory of universal quantum computation over continuous variables by combining linear optical devices such as phase shifters and beamsplitters with nonlinear elements such as downconverters and parametric amplifiers.

The primary reason for developing a theory of quantum information over continuous variables is a practical one. Most communication channels involve the exchange of bosons (such as photons or phonons), which correspond to quantized harmonic oscillators. Accordingly, the theory of quantum information over continuous variables is essential for understanding the physical limits of communication. Moreover, many important sensing and measurement techniques, including microscopy and radar, rely on bosonic systems. Quantum information over continuous variables is the key to understanding the limits of sensing and measurement and for devising methods to attain those limits. With this motivation, this work explores the fundamental limits of quantum communication and sensing and introduces techniques to achieve them, beginning with a review of continuous variable quantum information theory in the following chapter.

1.6 Scope of This Work

This subsection lists the major research areas in continuous variable quantum computing and quantum information, and situate the contributions of this work within them.

Major areas of research include:

1. **Channel Capacity of the Noisy, Lossy Bosonic Channel:** A fundamental question in quantum communication is: what is the maximum rate at which information can be sent down a bosonic channel in the presence of loss and noise? The capacity of a quantum channel $\mathcal{E}$ with inputs drawn from an ensemble of states $\hat{\rho}_k$ with probabilities p_k is given by its Holevo quantity,

$$\chi(\{p_k, \hat{\rho}_k\}) = S\left(\sum_k p_k \mathcal{E}(\hat{\rho}_k)\right) - \sum_k p_k S\left(\mathcal{E}(\hat{\rho}_k)\right), \tag{36}$$

where $S(\hat{\rho})$ is the von Neumann entropy of the state $\hat{\rho}$. Maximizing this quantity over all possible ensembles, including entangled inputs across multiple uses of the channel, yields the channel capacity. This maximization is often difficult to perform. The capacity of the bosonic channel with linear loss is known and is attained by sending ensembles of coherent states down the channel. By contrast, the capacity of the bosonic channel with linear loss and thermal noise remains unknown, although it is conjectured to also be attained by sending ensembles of coherent states.

2. **Minimum Output Entropy Conjecture:** The capacity of the bosonic channel with linear loss and thermal noise is indeed attained by coherent state inputs if the bosonic minimum output entropy conjecture holds. This conjecture states that a coherent-state input yields the output with the minimum von Neumann entropy for a bosonic channel with additive Gaussian noise. Proving or disproving the bosonic minimum output entropy conjecture is perhaps the most important open problem in continuous variable quantum information theory.

3. **Optimal Measurements for Distinguishing Bosonic States:** Distinguishing quantum states is an important component of quantum information science. Key results, such as the no-cloning theorem, the quantum teleportation protocol, and more recently the Yuen no-go theorem, have their origins in problems of state discrimination. The Helstrom bound gives the minimum probability of error technique for discriminating between two quantum states. The quantum Chernoff bound is an upper bound to the Helstrom limit of multiple-copy binary state discrimination, which is asymptotically tight in the error exponent. While the measurement techniques for attaining the Helstrom bound are known in principle, devising experimentally practical measurement techniques for attaining the Helstrom bound is not known in general. Moreover, even when the channel capacity for a quantum channel is attained by product state (unentangled) inputs, the measurement required to attain the capacity is the entangling 'pretty-good' measurement, which may be hard to perform in practice. For example, even though the channel capacity for the bosonic channel with linear loss is attained by coherent state inputs, practical decoding techniques that attain this bound are not currently known.

2 Continuous Variable Quantum Information Theory

2.1 Introduction

In Continuous Variable (CV) Quantum Information Theory, one goes beyond discrete systems (such as qubits) to systems described by **continuous degrees of freedom** . Many physical systems are naturally continuous. Examples include: Position and

momentum of a particle, electromagnetic field modes, vibrational modes of trapped ions. We traditionally formulate quantum information theory using qubits. Therefore, concepts in CV quantum information are often less familiar and requires - building a unified formalism, which is the main purpose of this review.
As we mentioned, many realistic quantum systems are continuous. A mode of the electromagnetic field behaves like a **quantum harmonic oscillator**. Studying the quantization of electromagnetic fields is important, as optical systems are widely used in communication and sensing, considering the experimental accessibility of photons. The theoretical framework we would describe here would be based on: the quantum harmonic oscillator, Creation and annihilation operators, various quantum gates and phase space representations (e.g., Wigner function). The tools built would allow us to describe quantum states in phase space, analyse Gaussian states, and study transformations via linear operations. Ultimately, the formalism developed here would be used for explaining bosonic quantum channels and state discrimination later.

2.2 Quantization of a Single Electromagnetic Field Mode

In this section, we will derive the Hamiltonian corresponding to a single mode of the electromagnetic field confined in a cavity. We will show that each mode behaves as a simple harmonic oscillator.

Consider a single mode of the electromagnetic field in a cavity of volume $V = AL$, where A is the cross-sectional area and L is the length along the z-direction.

We assume: The field propagates along the z-axis, the electric field is polarized along the x-direction, the magnetic field is along the y-direction.
The electric field is given by:

$$\mathbf{E}(z,t) = \hat{x}\sqrt{\frac{\omega^2}{4\epsilon_0 V}}\, q(t)\sin(kz) \tag{37}$$

The magnetic field is:

$$\mathbf{B}(z,t) = \hat{y}\sqrt{\frac{1}{4\epsilon_0 V}}\, \dot{q}(t)\cos(kz) \tag{38}$$

where:

- $q(t)$ is the generalized coordinate (field amplitude)
- $\dot{q}(t) = \frac{dq}{dt}$
- $\omega = ck$ is the angular frequency
- ϵ_0 is the permittivity of free space

2.2.1 Hamiltonian of the Field

The total energy of the electromagnetic field is:

$$H = \frac{1}{2}\int_V dV \left(\epsilon_0 E^2 + \frac{1}{\mu_0}B^2\right) \tag{39}$$

Substituting **E**:

$$E^2 = \frac{\omega^2}{4\epsilon_0 V} q^2(t)\sin^2(kz)$$

$$\epsilon_0 E^2 = \frac{\omega^2}{4V} q^2(t)\sin^2(kz)$$

Now for **B**:

$$B^2 = \frac{1}{4\epsilon_0 V}\dot{q}^2(t)\cos^2(kz)$$

Using $\frac{1}{\mu_0\epsilon_0} = c^2$:

$$\frac{1}{\mu_0}B^2 = \frac{c^2}{4V}\dot{q}^2(t)\cos^2(kz)$$

Since $\omega = ck$:

$$\frac{1}{\mu_0}B^2 = \frac{\omega^2}{4Vk^2}\dot{q}^2(t)\cos^2(kz)$$

Thus, we have the Total Hamiltonian in eq. (39) as:

$$H = \frac{1}{2}\int_V dV \left[\frac{\omega^2}{4V}q^2(t)\sin^2(kz) + \frac{c^2}{4V}\dot{q}^2(t)\cos^2(kz)\right]$$

Using:

$$\int_0^L \sin^2(kz)\,dz = \frac{L}{2}, \quad \int_0^L \cos^2(kz)\,dz = \frac{L}{2}$$

we obtain:

$$\int_V \sin^2(kz)\,dV = \frac{V}{2}, \quad \int_V \cos^2(kz)\,dV = \frac{V}{2}$$

Substituting:

$$H = \frac{1}{2}\left[\frac{\omega^2}{4V}q^2(t)\cdot\frac{V}{2} + \frac{c^2}{4V}\dot{q}^2(t)\cdot\frac{V}{2}\right]$$

Simplifying:

$$H = \frac{1}{2}\left[\frac{\omega^2}{8}q^2(t) + \frac{c^2}{8}\dot{q}^2(t)\right]$$

we rewrite:

$$H = \frac{1}{16}\left[\omega^2 q^2(t) + c^2 \dot{q}^2(t)\right] \tag{40}$$

We now absorb the constants into the variables by redefining:

$$q(t) \to \frac{\omega}{4\sqrt{\hbar\omega}}\, q, \qquad \dot{q}(t) \to \frac{c}{4\sqrt{\hbar\omega}}\, \dot{q}(t)$$

Substituting in eq. (40), we obtain:

$$H = \hbar\omega(q^2 + \dot{q}^2) = \hbar\omega(q^2 + p^2) \tag{41}$$

where $p = \dot{q}$

Thus, a single mode of the electromagnetic field behaves as a simple harmonic oscillator with frequency ω. This forms the basis for quantization.

2.3 Bosonic Modes from the Harmonic Oscillator

From the previous section, the Hamiltonian of a single electromagnetic field mode can be written as:

$$\hat{H} = \hbar\omega(\hat{q}^2 + \hat{p}^2)$$

where $\hat{q}$ and $\hat{p}$ are dimensionless quadrature operators.

2.3.1 Connection to the Standard Harmonic Oscillator

We now relate the dimensionless operators $(\hat{q}, \hat{p})$ to the physical position and momentum operators $(\hat{x}, \hat{p}_x)$ of the harmonic oscillator of mass m, with oscillation frequency ω.

Define:

$$\hat{q} = \sqrt{\frac{m\omega}{2\hbar}}\, \hat{x}, \qquad \hat{p} = \frac{1}{\sqrt{2m\hbar\omega}}\, \hat{p}_x \tag{42}$$

Substituting into the Hamiltonian in eq. (41):

$$\begin{aligned}\hat{H} &= \hbar\omega\left(\frac{m\omega}{2\hbar}\hat{x}^2 + \frac{1}{2m\hbar\omega}\hat{p}_x^2\right)\\ &= \frac{1}{2}m\omega^2\hat{x}^2 + \frac{\hat{p}_x^2}{2m}\end{aligned}$$

Thus, we recover the standard harmonic oscillator Hamiltonian:

$$\hat{H} = \frac{\hat{p}_x^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2$$

In quantum mechanics, the position and momentum operators satisfy:

$$[\hat{x}, \hat{p}_x] = i\hbar$$

Using the definitions of $\hat{q}$ and $\hat{p}$, we compute:

$$\begin{aligned}
[\hat{q}, \hat{p}] &= \left[\sqrt{\frac{m\omega}{2\hbar}}\hat{x}, \frac{1}{\sqrt{2m\hbar\omega}}\hat{p}_x\right] \\
&= \frac{1}{2\hbar}[\hat{x}, \hat{p}_x] \\
&= \frac{1}{2\hbar}(i\hbar) \\
&= \frac{i}{2}
\end{aligned}$$

Thus, the quadrature operators satisfy:

$$[\hat{q}, \hat{p}] = \frac{i}{2} \tag{43}$$

2.3.2 Creation and Annihilation Operators

We now define the bosonic ladder operators:

$$\hat{a} = \hat{q} + i\hat{p}, \qquad \hat{a}^\dagger = \hat{q} - i\hat{p}$$

They satisfy the commutation relation:

$$\begin{aligned}
[\hat{a}, \hat{a}^\dagger] &= [\hat{q} + i\hat{p}, \hat{q} - i\hat{p}] \\
&= -i[\hat{q}, \hat{p}] + i[\hat{p}, \hat{q}] \\
&= -i\left(\frac{i}{2}\right) + i\left(-\frac{i}{2}\right) \\
&= 1
\end{aligned}$$

2.3.3 Hamiltonian in Terms of Ladder Operators

Using:

$$\begin{aligned}
\hat{a}^\dagger\hat{a} &= (\hat{q} - i\hat{p})(\hat{q} + i\hat{p}) \\
&= \hat{q}^2 + i\hat{q}\hat{p} - i\hat{p}\hat{q} - i^2\hat{p}^2 \\
&= \hat{q}^2 + \hat{p}^2 + i(\hat{q}\hat{p} - \hat{p}\hat{q}) \\
&= \hat{q}^2 + \hat{p}^2 + i[\hat{q}, \hat{p}] \\
&= \hat{q}^2 + \hat{p}^2 - \frac{1}{2}
\end{aligned}$$

we obtain:

$$\begin{aligned}\hat{H} &= \hbar\omega(\hat{q}^2 + \hat{p}^2) \\ &= \hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right)\end{aligned}$$

Physical Interpretation: The operators $\hat{a}$ and $\hat{a}^\dagger$ describe annihilation and creation of quanta of the electromagnetic field. Each mode behaves as a quantum harmonic oscillator, and its energy eigenstates correspond to photon number states.

2.3.4 Fock States and Vacuum State

We define the vacuum state $|0\rangle$ as:

$$\hat{a}\,|0\rangle = 0 \tag{44}$$

Higher number states are generated by repeated application of the creation operator:

$$|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}\,|0\rangle \tag{45}$$

These states satisfy:

$$\hat{N}\,|n\rangle = n\,|n\rangle \tag{46}$$

where we have defined the number operator as: $\hat{N} = \hat{a}\hat{a}^\dagger$ The annihilation and creation operators act as:

$$\hat{a}\,|n\rangle = \sqrt{n}\,|n-1\rangle\,, \quad n \geq 1$$

$$\hat{a}^\dagger\,|n\rangle = \sqrt{n+1}\,|n+1\rangle$$

Thus, the Fock states represent states with a definite number of quanta (photons).

2.3.5 Multi-Mode Bosonic System

We now generalize to a system of n independent modes.

Each mode is described by its own annihilation operator:

$$\hat{a}_k, \quad k = 1, 2, \ldots, n$$

These operators satisfy the canonical commutation relations:

$$[\hat{a}_k, \hat{a}_l^\dagger] = \delta_{kl} \tag{47}$$

The Hilbert space of the system is the tensor product of the individual Fock spaces'

hamiltonians:

$$\mathcal{H} = \bigotimes_{k=1}^{n} \mathcal{H}_k \tag{48}$$

The number operator for each mode is:

$$\hat{N}_k = \hat{a}_k^\dagger \hat{a}_k \tag{49}$$

The total Hamiltonian of the system is:

$$\hat{H} = \sum_{k=1}^{n} \hbar\omega_k \left(\hat{a}_k^\dagger \hat{a}_k + \frac{1}{2} \right) \tag{50}$$

2.3.6 Quadrature Operators for Each Mode

For each mode, we define the quadrature operators:

$$\hat{q}_k = \frac{1}{2}(\hat{a}_k + \hat{a}_k^\dagger)$$

$$\hat{p}_k = \frac{1}{2i}(\hat{a}_k - \hat{a}_k^\dagger)$$

These satisfy the commutation relation:

$$[\hat{q}_k, \hat{p}_l] = \frac{i}{2}\delta_{kl}$$

2.3.7 Phase Space Representation

We now introduce the vector of quadrature operators:

$$\hat{R} = (\hat{q}_1, \hat{p}_1, \hat{q}_2, \hat{p}_2, \ldots, \hat{q}_n, \hat{p}_n)^T$$

The commutation relations can be compactly written as:

$$[\hat{R}_i, \hat{R}_j] = \frac{i}{2}\Omega_{ij} \tag{51}$$

where Ω is the symplectic form:

$$\Omega = \bigoplus_{k=1}^{n} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \tag{52}$$

The symplectic form Ω can be written explicitly as a $2n \times 2n$ matrix:

$$\Omega = \begin{pmatrix} 0 & 1 & 0 & 0 & \cdots & 0 \\ -1 & 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 1 & \cdots & 0 \\ 0 & 0 & -1 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 1 \\ 0 & 0 & 0 & 0 & \cdots & -1 \end{pmatrix} \tag{53}$$

This matrix encodes the canonical commutation relations between all quadrature operators and defines the symplectic structure of the phase space.

2.4 Mean Vector, Expectation Values and the Covariance Matrix

We now complete the phase-space description by defining expectation values.

For a quantum system described by a density operator $\hat{\rho}$, the expectation value of any operator $\hat{O}$ is defined as:

$$\langle \hat{O} \rangle = \mathrm{tr}(\hat{\rho}\hat{O}) \tag{54}$$

Here, the density operator $\hat{\rho}$ describes the full state of the system, which may be either pure or mixed.

The mean vector (first moments) of the quadrature operators is defined as:

$$\bar{R}_k = \langle \hat{R}_k \rangle = \mathrm{tr}(\hat{\rho}\hat{R}_k) \tag{55}$$

Thus, $\bar{R}$ gives the average values of all quadrature operators and represents the displacement of the state in phase space.

2.4.1 Covariance Matrix

The second moments of the quadrature operators are captured by the covariance matrix σ, defined as:

$$\sigma_{kl} = \frac{1}{2}\langle \hat{R}_k\hat{R}_l + \hat{R}_l\hat{R}_k \rangle - \langle \hat{R}_k \rangle \langle \hat{R}_l \rangle \tag{56}$$

Using the anticommutator $\{A, B\} = AB + BA$, this can be written as:

$$\sigma_{kl} = \frac{1}{2}\langle \{\hat{R}_k, \hat{R}_l\} \rangle - \langle \hat{R}_k \rangle \langle \hat{R}_l \rangle \tag{57}$$

The covariance matrix describes: the uncertainties (variances) of quadrature operators and the correlations between different modes.

Gaussian States

A quantum state is said to be Gaussian if its phase-space representation (such as the Wigner function) is Gaussian.

In this case, the full quantum state is completely characterized by: the mean vector $\bar{R}$, and the covariance matrix σ.

When multiple modes are considered, the state is referred to as a joint Gaussian state, meaning that the joint phase-space distribution of all quadratures is a multivariate Gaussian.

Physical Significance

The mean vector determines the position of the state in phase space, while the covariance matrix determines its spread and correlations.

These quantities will become relevant in the study of Gaussian states, Gaussian transformations, and bosonic quantum channels.

The symplectic structure defined by Ω is further explored in the following section:

2.5 The Symplectic Structure of Phase Space and the Group $\mathrm{Sp}(2n, \mathbb{R})$

In the previous section, we introduced the vector of quadrature operators:

$$\hat{R} = (\hat{q}_1, \hat{p}_1, \hat{q}_2, \hat{p}_2, \ldots, \hat{q}_n, \hat{p}_n)^T$$

These operators satisfy the canonical commutation relations:

$$[\hat{R}_i, \hat{R}_j] = \frac{i}{2}\Omega_{ij}$$

where Ω is the **symplectic form**, given by:

$$\Omega = \begin{pmatrix} 0 & 1 & 0 & 0 & \cdots & 0 \\ -1 & 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 1 & \cdots & 0 \\ 0 & 0 & -1 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & 1 \\ 0 & 0 & 0 & 0 & -1 & 0 \end{pmatrix} = \bigoplus_{k=1}^{n} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

2.5.1 Definition of Symplectic Transformations

Let us define the set of all real $2n \times 2n$ matrices:

$$\mathcal{M}(2n) = \{S \mid S \text{ is a real } 2n \times 2n \text{ matrix}\}$$

A matrix $S \in \mathcal{M}(2n)$ is called **symplectic** if it satisfies:

$$S^T \Omega S = \Omega \tag{58}$$

This condition ensures that the transformation preserves the symplectic structure of phase space.

2.5.2 The Symplectic Group

The set of all matrices satisfying Eq. (58) forms a group under matrix multiplication, called the **real symplectic group**:

$$\mathrm{Sp}(2n, \mathbb{R}) = \{S \in \mathcal{M}(2n) \mid S^T \Omega S = \Omega\}$$

This group has the following properties:

- Closure: If $S_1, S_2 \in \mathrm{Sp}(2n, \mathbb{R})$, then $S_1 S_2 \in \mathrm{Sp}(2n, \mathbb{R})$
- Identity: The identity matrix I belongs to $\mathrm{Sp}(2n, \mathbb{R})$
- Inverse: If $S \in \mathrm{Sp}(2n, \mathbb{R})$, then $S^{-1} \in \mathrm{Sp}(2n, \mathbb{R})$
- Associativity: Follows from matrix multiplication

2.5.3 Real Canonical Linear Transformations

Consider a real canonical linear transformation (where quadratures obey the canonical commutation relation in eq. (51) of the quadrature operators:

$$\hat{R}' = S\hat{R} \tag{59}$$

We now compute the commutation relations of the transformed operators:

$$\begin{aligned}
[\hat{R}'_i, \hat{R}'_j] &= \left[\sum_k S_{ik}\hat{R}_k, \sum_l S_{jl}\hat{R}_l\right] \\
&= \sum_{k,l} S_{ik}S_{jl}[\hat{R}_k, \hat{R}_l] \\
&= \sum_{k,l} S_{ik}S_{jl}\left(\frac{i}{2}\Omega_{kl}\right) \\
&= \frac{i}{2}(S\Omega S^T)_{ij}
\end{aligned}$$

As it is a real canonical linear transformation, for the commutation relations to remain invariant, we require:

$$S\Omega S^T = \Omega$$

Taking transpose of both sides:

$$S^T\Omega S = \Omega \tag{60}$$

which is precisely the symplectic condition.

Thus, for every canonical linear transformation S, we have S $\in$ Sp$(2n, \mathbb{R})$. Conversely for any S $\in$ Sp$(2n, \mathbb{R})$, we can have a canonical linear transformation $\hat{R}' = S\hat{R}$. Thus, **real canonical linear transformations in 2n dimensions** and **real 2n x 2n symplectic matrices** have a one-to-one correspondence. Symplectic transformations (S $\in$ Sp$(2n, \mathbb{R})$) are important because: they preserve the **uncertainty relations** in quantum mechanics, and they will describe (as we will see later) **Gaussian unitary operations** in continuous-variable quantum systems.

2.6 Quadratic Hamiltonians and Gaussian Transformations

We study Hamiltonians that are at most quadratic in the quadrature operators $\hat{R} = (\hat{R}_1, \ldots, \hat{R}_{2n})^T$. The general form is:

$$\hat{H} = \frac{1}{2}\hat{R}^T G\hat{R} + \mathbf{b}^T\hat{R}, \tag{61}$$

where G is a real, symmetric $2n \times 2n$ matrix ($G = G^T$) and $\mathbf{b}$ is a real vector. The symmetry of G ensures Hermiticity of our Hamiltonian. Such quadratic Hamiltonians also generate *Gaussian transformations*, mapping Gaussian states to Gaussian states.

Heisenberg Evolution: The time evolution of the quadrature operators is governed by the Heisenberg equation. Using the canonical commutation relations $[\hat{R}_i, \hat{R}_j] = \frac{i}{2}\Omega_{ij}$

(eq. (51)), we derive the equation of motion for $\hat{R}$.

Since the quadrature operators $\hat{R}_m$ have no explicit time dependence, the equation of motion for the m-th component is:

$$\frac{d\hat{R_m}}{dt} = \frac{i}{\hbar}[\hat{H}, \hat{R_m}]. \tag{62}$$

First, consider the commutator of the quadratic term of the Hamiltonian $\hat{H} = \frac{1}{2}\hat{R}^T G\hat{R}$ $= \frac{1}{2}\sum_{k,l} G_{kl}\hat{R}_k\hat{R}_l$ with $\hat{R}_m$:

$$\begin{aligned}\left[\frac{1}{2}\sum_{k,l} G_{kl}\hat{R}_k\hat{R}_l, \hat{R}_m\right] &= \frac{1}{2}\sum_{k,l} G_{kl}\left(\hat{R}_k[\hat{R}_l, \hat{R}_m] + [\hat{R}_k, \hat{R}_m]\hat{R}_l\right) \\ &= \frac{i}{4}\sum_{k,l} G_{kl}\left(\hat{R}_k\Omega_{lm} + \Omega_{km}\hat{R}_l\right).\end{aligned} \tag{63}$$

By exploiting the symmetry of G and antisymmetry of Ω ($\Omega^T = -\Omega$), both terms combine into a single matrix product. Specifically, first term in eq. (63) $\sum_{k,l} G_{kl}\Omega_{lm}\hat{R}_k = \sum_{k,l}((G\Omega)^T)_{mk}\hat{R}_k = -\sum_k(\Omega G)_{mk}\hat{R}_k$ (as $\Omega^T = -\Omega$ and $G^T = G$) and similarly for the second term in eq. (63). Thus:

$$[\hat{H}_{quad}, \hat{R}] = -\frac{i}{2}\Omega G\hat{R}. \tag{64}$$

Now, for the linear term in H, $[\mathbf{b}^T\hat{R}, \hat{R}_m] = \sum_k b_k\frac{i}{2}\Omega_{km} = -\sum_k \frac{i}{2}\Omega_{mk}b_k = -\frac{i}{2}(\Omega\mathbf{b})_m$.

Combining these results and applying the Heisenberg equation $\frac{d\hat{R}}{dt} = \frac{i}{\hbar}[\hat{H}, \hat{R}]$ (where constants are absorbed into the definition of Ω), we obtain the linear differential equation:

$$\frac{d\hat{R}}{dt} = \Omega G\hat{R} + \Omega\mathbf{b}. \tag{65}$$

Equation 65 is a linear differential equation. Defining $A = \Omega G$ and $\mathbf{c} = \Omega\mathbf{b}$, the formal solution is:

$$\hat{R}(t) = S(t)\hat{R}(0) + \mathbf{d}(t), \tag{66}$$

where $S(t) = e^{At} = e^{\Omega Gt}$ is the homogeneous solution and $\mathbf{d}(t) = \int_0^t e^{\Omega G(t-s)}\Omega\mathbf{b}\,ds$ is the displacement.

Symplectic Nature of $S(t)$: We verify that $S(t)$ is a symplectic transformation, i.e., $S(t)^T\Omega S(t) = \Omega$. Differentiating with respect to time:

$$\frac{d}{dt}\left(S^T\Omega S\right) = S^T\left(A^T\Omega + \Omega A\right)S. \tag{67}$$

Substituting $A = \Omega G$ and using $G^T = G, \Omega^T = -\Omega$:

$$\begin{aligned} A^T\Omega + \Omega A &= (\Omega G)^T\Omega + \Omega(\Omega G) \\ &= G^T\Omega^T\Omega + \Omega^2 G \\ &= G(-\Omega)\Omega + \Omega^2 G \\ &= -G\Omega^2 + \Omega^2 G \\ &= -G(-I) + (-I)G \quad (\text{since } \Omega^2 = -I) \\ &= G - G = 0. \end{aligned} \tag{68}$$

Since the time derivative is zero, the quantity $S(t)^T\Omega S(t)$ is constant in time. Evaluating at $t = 0$ where $S(0) = I$:

$$S(t)^T\Omega S(t) = S(0)^T\Omega S(0) = I^T\Omega I = \Omega.$$

Thus, $S(t)$ satisfies the definition of a symplectic matrix for all t.

Transformation of Mean and Covariance Matrix: The mean vector is defined as:

$$\bar{R} = \langle \hat{R} \rangle \tag{69}$$

From eq. (66):

$$\bar{R} \to S\bar{R} + \mathbf{d} \tag{70}$$

The covariance matrix is:

$$\sigma_{ij} = \frac{1}{2}\langle \{\hat{R}_i - \bar{R}_i, \hat{R}_j - \bar{R}_j\} \rangle \tag{71}$$

Now,

$$\hat{R}' - \bar{R}' = S(\hat{R} - \bar{R}). \tag{72}$$

Substituting into the definition of the covariance matrix,

$$\begin{aligned} \sigma'_{ij} &= \frac{1}{2}\left\langle \left\{ \hat{R}'_i - \bar{R}'_i, \hat{R}'_j - \bar{R}'_j \right\} \right\rangle \\ &= \frac{1}{2}\left\langle \left\{ (S(\hat{R} - \bar{R}))_i, (S(\hat{R} - \bar{R}))_j \right\} \right\rangle. \end{aligned}$$

Using linearity,

$$(S(\hat{R} - \bar{R}))_i = \sum_k S_{ik}(\hat{R}_k - \bar{R}_k), \tag{73}$$

we obtain

$$\begin{aligned}\sigma'_{ij} &= \sum_{k,l} S_{ik} S_{jl} \frac{1}{2} \left\langle \left\{ \hat{R}_k - \bar{R}_k, \hat{R}_l - \bar{R}_l \right\} \right\rangle \\ &= \sum_{k,l} S_{ik} \sigma_{kl} S_{jl}.\end{aligned}$$

Hence, in matrix form,

$$\sigma \to \sigma' = S\sigma S^T. \tag{74}$$

Under such transformations:

$$\sigma \to S\sigma S^T \tag{75}$$

Now, we see how **Gaussian states** are transformed:

The Wigner function for Gaussian states is:

$$W(R) = \frac{\exp\left[-\frac{1}{2}(R-\bar{R})^T \sigma^{-1} (R-\bar{R})\right]}{(2\pi)^N \sqrt{\det \sigma}} \tag{76}$$

With $R' = SR + d$, we invert to $R = S^{-1}(R' - d)$. The exponent transforms as:

$$\begin{aligned}(R-\bar{R})^T \sigma^{-1} (R-\bar{R}) &= (R'-\bar{R}')^T (S^{-1})^T \sigma^{-1} S^{-1} (R'-\bar{R}') \\ &= (R'-\bar{R}')^T (S\sigma S^T)^{-1} (R'-\bar{R}')\end{aligned}$$

where $\bar{R}' = S\bar{R} + d$ and $\sigma' = S\sigma S^T$. Since $\det S = 1$, $\det \sigma' = \det \sigma$. Thus:

$$W'(R') = \frac{\exp\left[-\frac{1}{2}(R'-\bar{R}')^T (\sigma')^{-1} (R'-\bar{R}')\right]}{(2\pi)^N \sqrt{\det \sigma'}} \tag{77}$$

Thus, Gaussian states are mapped to Gaussian states, and such unitary transformations are also called **Gaussian transformations**.

2.7 Characteristic Functions (Quantum)

The characteristic function of a quantum state ρ is defined as

$$\chi(\alpha) = \mathrm{Tr}\left[\rho D(\alpha)\right], \quad D(\alpha) = \exp\left(\alpha a^\dagger - \alpha^* a\right) \tag{78}$$

(for displacement operator $D(\alpha)$ refer to eq. (100)) Because a and $a^\dagger$ do not commute, different operator orderings give different characteristic functions [**braunstein2005cv**],

which correspond to different quasi-probability distributions in phase space::

$$\chi(\alpha) = \mathrm{Tr}[\rho D(\alpha)] \quad \text{(Weyl / symmetric)}, \tag{79}$$

$$\chi_N(\alpha) = \mathrm{Tr}\left[\rho e^{\alpha a^\dagger} e^{-\alpha^* a}\right] \quad \text{(normal ordering)}, \tag{80}$$

$$\chi_A(\alpha) = \mathrm{Tr}\left[\rho e^{-\alpha^* a} e^{\alpha a^\dagger}\right] \quad \text{(anti-normal ordering)}. \tag{81}$$

These correspond respectively to different phase-space distributions (Wigner, P, and Q functions).

For Gaussian states,$\chi(\xi) = \exp\left(-\frac{1}{2}\xi^T \sigma \xi + i\,\xi^T \Omega \bar{R}\right)$

Here $\xi \in \mathbb{R}^{2n}$ denotes the real phase-space vector corresponding to the complex displacement α. So, the state is fully determined by mean $\bar{R}$ and covariance σ.

Relation to Classical Probability:

In classical theory, the characteristic function of a random variable x is

$$\chi_{ct}(\lambda) = \mathbb{E}[e^{i\lambda x}] = \int p(x) e^{i\lambda x} dx. \tag{82}$$

The quantum version is a direct analogue: as if we are finding expectation value ($\mathrm{Tr}[\rho D(\alpha)]$) of $D(\alpha)$ (like of $e^{i\lambda x}$ in classical theory).

2.8 Wigner Function

The Wigner function provides a phase-space representation of a quantum state [**weedbrook2012gaussian**
For a single-mode density operator $\hat{\rho}$, it is defined as

$$W(q,p) = \frac{1}{\pi\hbar} \int_{-\infty}^{\infty} dy\, \langle q - y|\hat{\rho}|q + y\rangle\, e^{2ipy/\hbar} \tag{83}$$

Characteristic Function Representation: The Wigner function provides an equivalent description of an n-mode quantum state in phase space to the characteristic function. It can be defined as the Fourier transform of the symmetrically ordered characteristic function $\chi(\boldsymbol{\lambda})$:

$$W(\boldsymbol{\alpha}) = \int \frac{d^{2n}\boldsymbol{\lambda}}{\pi^{2n}}\, \chi(\boldsymbol{\lambda})\, \exp\!\left(\boldsymbol{\alpha}\boldsymbol{\lambda}^\dagger - \boldsymbol{\alpha}^\dagger\boldsymbol{\lambda}\right) \tag{84}$$

For a single mode, this reduces to

$$W(\alpha) = \int \frac{d^{2}\boldsymbol{\lambda}}{\pi^{2}}\, \chi(\boldsymbol{\lambda})\, \exp\!\left(\boldsymbol{\alpha}\boldsymbol{\lambda}^\dagger - \boldsymbol{\alpha}^\dagger\boldsymbol{\lambda}\right) \tag{85}$$

Here, $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_n)^T$ is the vector of complex phase-space coordinates, where each component is related to the quadrature operators by $\alpha_k = (q_k + ip_k)/\sqrt{2}$. The variable $\boldsymbol{\lambda}$

is the integration variable in the Fourier transform, corresponding to the argument of the displacement operator in the characteristic function $\chi(\boldsymbol{\lambda}) = \mathrm{Tr}[\hat{\rho}\hat{D}(\boldsymbol{\lambda})]$.

Properties:

- Normalization: $\int dq\, dp\, W(q,p) = 1$.
- $W(q,p)$ is real, but not necessarily positive.
- Marginals:
$$\int dp\, W(q,p) = \langle q|\hat{\rho}|q\rangle, \quad \int dq\, W(q,p) = \langle p|\hat{\rho}|p\rangle.$$

Expectation Values:
$$\langle \hat{A}\rangle = \int d^{2n}\mathbf{R}\, W_\rho(\mathbf{R})\, A_W(\mathbf{R}), \tag{86}$$
where $A_W(\mathbf{R})$ is the Weyl symbol of $\hat{A}$.

2.9 P Representation

The Glauber–Sudarshan P-distribution provides a diagonal representation of the density operator in the coherent-state basis:
$$\hat{\rho} = \int d^2\alpha\, P(\alpha)\, |\alpha\rangle\langle\alpha|. \tag{87}$$

Properties:

- **Diagonal form:** Expresses $\hat{\rho}$ as a weighted mixture of coherent states.
- **Classicality:** If $P(\alpha)$ behaves like a true probability distribution (positive and regular), the state is classical.
- **Non-classicality:** For many quantum states (e.g., Fock states), $P(\alpha)$ becomes highly singular or non-positive.
- **Characteristic Function:** Related to the normally ordered characteristic function:
$$\chi_N(\lambda) = \int d^2\alpha\, P(\alpha)\, e^{\lambda\alpha^* - \lambda^*\alpha}. \tag{88}$$

2.10 Trace Rules in Phase Space

Trace identities allow operator expressions to be evaluated as integrals in phase space, which are often simpler to compute.

Glauber Expansion: Any operator $\hat{O}$ acting on an n-mode Hilbert space can be written as

$$\hat{O} = \int \frac{d^{2n}\lambda}{\pi^n} \chi[\hat{O}](\lambda)\, \hat{D}(-\lambda), \tag{89}$$

where $\chi[\hat{O}](\lambda)$ is the characteristic function.

Trace Rule (Characteristic Functions): For two operators $\hat{O}_1$ and $\hat{O}_2$:

$$\mathrm{Tr}(\hat{O}_1\hat{O}_2) = \int \frac{d^{2n}\lambda}{\pi^n} \chi[\hat{O}_1](\lambda)\, \chi[\hat{O}_2](-\lambda). \tag{90}$$

Trace Rule (Wigner Functions): Equivalently, in terms of Wigner functions:

$$\mathrm{Tr}(\hat{O}_1\hat{O}_2) = (\pi)^n \int d^{2n}\lambda\, W[\hat{O}_1](\lambda)\, W[\hat{O}_2](\lambda). \tag{91}$$

which will be used to derive the fidelity and Quantum Chernoff Bound for Gaussian states, in later sections.

2.11 Uncertainty Principle

The uncertainty principle follows from the non-commutativity of operators. For any two observables A and B, the Robertson uncertainty relation is given by

$$\Delta A \Delta B \geq \frac{1}{2} \left|\langle [A, B] \rangle\right| \tag{92}$$

For continuous variable systems, the canonical quadrature operators $\hat{q}$ and $\hat{p}$ satisfy

$$[\hat{q}, \hat{p}] = i\hbar, \tag{93}$$

which leads to the standard uncertainty relation

$$\Delta q \Delta p \geq \frac{\hbar}{2} \tag{94}$$

More generally, for an n-mode system described by the covariance matrix σ, the uncertainty principle can be written in compact form [**simon1994quantum**] as

$$\sigma + \frac{i\hbar}{2}\Omega \geq 0 \tag{95}$$

where Ω is the symplectic form.

2.12 Coherent States

2.12.1 Fock representation of the coherent state

Assume that the annihilation operator $\hat{a}$ has a right eigenstate. Because $\hat{a}$ is not Hermitian, its eigenvalue will in general be some complex number α, and we label the corresponding eigenstate $|\alpha\rangle$ by the same complex number α. We may therefore write the equation

$$\hat{a}|\alpha\rangle = \alpha|\alpha\rangle.$$

The conjugate state $\langle\alpha|$ is obviously a left eigenstate of the creation operator $\hat{a}^\dagger$, as can be seen by taking the Hermitian conjugate.

$$\langle\alpha|\hat{a}^\dagger = \alpha^*\langle\alpha|.$$

The state $|\alpha\rangle$ will be called the coherent state. Our initial goal is to find an explicit representation for $|\alpha\rangle$ in the basis of Fock states $|n\rangle$ labelled by the occupation number n. Since the Fock states form a complete set, we may use them to represent $|\alpha\rangle$ in the form

$$|\alpha\rangle = \sum_{n=0}^{\infty} c_n|n\rangle,$$

in which the c_n are complex numbers to be determined. On substituting the expansion and using the "step down" equation, we obtain

$$\sum_{n=1}^{\infty} c_n\sqrt{n}|n-1\rangle = \alpha\sum_{n=0}^{\infty} c_n|n\rangle.$$

As the $|n\rangle$ $(n = 0, 1, 2, ...)$ are a set of orthogonal state vectors, this equation is satisfied only if the coefficients of corresponding Fock state vectors on both sides are equal. We therefore have, on equating coefficients of $|n-1\rangle$,

$$c_n = \frac{\alpha}{\sqrt{n}}c_{n-1}.$$

This is a recursion formula connecting c_n and c_{n-1} from which we obtain by repeated application

$$c_n = \frac{\alpha^2}{[n(n-1)]^{1/2}}c_{n-2} = ... = \frac{\alpha^n}{\sqrt{n!}}c_0,$$

so that

$$|\alpha\rangle = c_0\sum_{n=0}^{\infty}\frac{\alpha^n}{\sqrt{n!}}|n\rangle.$$

c_0 can be determined from the requirement that the state $|\alpha\rangle$ be normalized to unity,

which implies

$$\begin{aligned}\langle\alpha|\alpha\rangle = 1 = |c_0|^2 \sum_{n=0}^{\infty}\sum_{m=0}^{\infty} \frac{\alpha^{*n}\alpha^m}{\sqrt{n!}\sqrt{m!}}\langle n|m\rangle \\ = |c_0|^2 \sum_{n=0}^{\infty} \frac{|\alpha|^{2n}}{n!} \\ = |c_0|^2 e^{|\alpha|^2},\end{aligned}$$

when we make use of the orthogonality of the Fock states. Hence

$$|c_0| = e^{-|\alpha|^2/2},$$

and, apart from a unimodular factor, we have

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}}|n\rangle, \tag{96}$$

and

$$\langle\alpha| = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^{*n}}{\sqrt{n!}}\langle n|.$$

It is interesting to note that for every complex number α, other than zero, the coherent state $|\alpha\rangle$ has a non-zero projection on every Fock state $|n\rangle$. Thus

$$\langle n|\alpha\rangle = e^{-|\alpha|^2/2}\frac{\alpha^n}{\sqrt{n!}}.$$

When $\alpha = 0$, the coherent state $|\alpha\rangle$ becomes the vacuum state $|vac\rangle$, which may be regarded as either a coherent state or a Fock state. The squared modulus of the projection of $|\alpha\rangle$ onto $|n\rangle$ gives the probability $p(n)$ that n excitations or photons will be found in the coherent state $|\alpha\rangle$. We therefore have

$$p(n) = |\langle n|\alpha\rangle|^2 = \frac{|\alpha|^{2n}}{n!}e^{-|\alpha|^2}, \tag{97}$$

which will be recognized as a Poisson distribution in n, with parameter $|\alpha|^2$. The mean number of photons present when the state is a coherent state $|\alpha\rangle$ is therefore given by(Poisson distbn - mean is the λ of the distbn. - here $|\alpha|^2$).

$$\sum_{n=0}^{\infty} np(n) = |\alpha|^2 = \langle\alpha|\hat{a}^\dagger\hat{a}|\alpha\rangle \tag{98}$$

and this is large or small according as α is a large or small number.

But, no matter how small α may be (with the exception of $\alpha = 0$), there is some

non-zero probability $p(n)$ that any number of photons n, however large, is present in the field. In a certain sense the photon number is as random as possible in a coherent state, subject to a certain mean number $|\alpha|^2$. It is a remarkable feature of the combination of Fock states that the state is unchanged when acted on by the annihilation operator $\hat{a}$. A possible connection between the coherent state of the quantum field and a classical field is suggested by the fact that it is possible to absorb photons from an electromagnetic field in a coherent state repeatedly, without changing the state in any way.

At first sight it might seem that, because the coherent state is not the eigenstate of any observable, it does not correspond to any readily measurable feature of the electromagnetic field. However, this is not so, because in practice most measurements of the field in the optical domain are based on the process of photoelectric detection. This includes instruments such as the photomultiplier, the photoconductor, the photographic plate and the eye. These instruments function by the absorption of photons, and it is therefore the absorption operator which is most closely associated with the measurement of the field. It is partly because the coherent states are eigenstates of the absorption operator, that these states prove to be particularly convenient for the description of many properties of the field encountered in photoelectric measurements.

2.12.2 The coherent state as a displaced vacuum state - the displacement operator

By combining the expansion with the representation of the Fock state $|n\rangle = \frac{\hat{a}^{\dagger n}}{\sqrt{n!}}|vac\rangle$, we can write

$$\begin{aligned}|\alpha\rangle &= e^{-|\alpha|^2/2}\sum_{n=0}^{\infty}\frac{\alpha^n\hat{a}^{\dagger n}}{n!}|vac\rangle \\ &= e^{-|\alpha|^2/2}e^{\alpha\hat{a}^\dagger}|vac\rangle,\end{aligned}$$

which shows that the coherent state $|\alpha\rangle$ may also be regarded as a displaced vacuum state .

We can re-express this in a somewhat more symmetric form, by inserting the operator $e^{-\alpha^*\hat{a}}$ between $e^{\alpha\hat{a}^\dagger}$ and $|vac\rangle$. Thus

$$|\alpha\rangle = e^{-|\alpha|^2/2}e^{\alpha\hat{a}^\dagger}e^{-\alpha^*\hat{a}}|vac\rangle. \tag{99}$$

The insertion of $e^{-\alpha^*\hat{a}}$ can be justified by expansion of the exponential operator, which leads to the result

$$\begin{aligned}e^{-\alpha^*\hat{a}}|vac\rangle &= \left[1-\alpha^*\hat{a}+\frac{(\alpha^*\hat{a})^2}{2!}-...\right]|vac\rangle \\ &= |vac\rangle.\end{aligned}$$

We now make use of the Campbell-Baker-Hausdorff operator identity for two operators $\hat{A}$, $\hat{B}$ provided that

$$\exp\Big(\hat{A}+\hat{B}\Big) = \exp\hat{A}\exp\hat{B}\exp\Big(-[\hat{A},\hat{B}]/2\Big),$$

$$[\hat{A},[\hat{A},\hat{B}]] = 0 = [\hat{B},[\hat{A},\hat{B}]].$$

The condition is obviously satisfied for any pair of operators $\hat{A}$, $\hat{B}$ whose commutator $[\hat{A},\hat{B}]$ is a c-number. If we now put $\hat{A} = \alpha\hat{a}^\dagger$, $\hat{B} = -\alpha^*\hat{a}$ and make use of the commutation relation, we obtain $[\hat{A},\hat{B}] = |\alpha|^2$ and

$$e^{\alpha\hat{a}^\dagger - \alpha^*\hat{a}} = e^{\alpha\hat{a}^\dagger}e^{-\alpha^*\hat{a}}e^{-|\alpha|^2/2}$$

which allows us to re-write the equation in the more compact form

$$|\alpha\rangle = \hat{D}(\alpha)|0\rangle, \tag{100}$$

where

$$\hat{D}(\alpha) \equiv e^{\alpha\hat{a}^\dagger - \alpha^*\hat{a}}.$$

$\hat{D}(\alpha)$ is the displacement operator that creates the coherent state $|\alpha\rangle$ from the vacuum state $|0\rangle = |vac\rangle$. It is clear by inspection that $\hat{D}(\alpha)$ is a unitary operator such that

$$\hat{D}^\dagger(\alpha)\hat{D}(\alpha) = 1 = \hat{D}(\alpha)\hat{D}^\dagger(\alpha),$$

and

$$\hat{D}^\dagger(\alpha) = \hat{D}(-\alpha).$$

The displacement operator also has the following property:

Unitary transformation of $\hat{a}$ or $\hat{a}^\dagger$ leads to the complex displacement α or α^*, respectively. Thus

$$\hat{D}^\dagger(\alpha)\hat{a}\hat{D}(\alpha) = \hat{a} + \alpha$$

and

$$\hat{D}^\dagger(\alpha)\hat{a}^\dagger\hat{D}(\alpha) = \hat{a}^\dagger + \alpha^*.$$

These properties follow by applying the operator expansion theorem (the Hadamard lemma), which states that for any operators $\hat{A}$ and $\hat{B}$:

$$e^{\hat{A}}\hat{B}e^{-\hat{A}} = \hat{B} + [\hat{A},\hat{B}] + \frac{1}{2!}[\hat{A},[\hat{A},\hat{B}]] + \dots$$

Letting the displacement operator be $\hat{D}(\alpha) = e^{\alpha\hat{a}^\dagger - \alpha^*\hat{a}}$, its adjoint is $\hat{D}^\dagger(\alpha) = e^{-\alpha\hat{a}^\dagger + \alpha^*\hat{a}}$. To map this to the theorem, we set $\hat{A} = -\alpha\hat{a}^\dagger + \alpha^*\hat{a}$. First, we evaluate the transformation

of $\hat{B} = \hat{a}$. Using the fundamental commutation relations $[\hat{a}, \hat{a}^\dagger] = 1$ and $[\hat{a}^\dagger, \hat{a}] = -1$, we compute the first-order commutator:

$$\begin{aligned}
[\hat{A}, \hat{a}] &= [-\alpha\hat{a}^\dagger + \alpha^*\hat{a}, \hat{a}] \\
&= -\alpha[\hat{a}^\dagger, \hat{a}] + \alpha^*[\hat{a}, \hat{a}] \\
&= -\alpha(-1) + 0 \\
&= \alpha
\end{aligned}$$

Because α is a c-number (scalar), it commutes with all operators. Therefore, all higher-order nested commutators strictly vanish:

$$[\hat{A}, [\hat{A}, \hat{a}]] = [\hat{A}, \alpha] = 0$$

Substituting these terms back into the expansion theorem, the infinite series truncates exactly after the second term:

$$\hat{D}^\dagger(\alpha)\hat{a}\hat{D}(\alpha) = e^{-\alpha\hat{a}^\dagger + \alpha^*\hat{a}}\hat{a}e^{\alpha\hat{a}^\dagger - \alpha^*\hat{a}} = \hat{a} + \alpha$$

We repeat the process for $\hat{B} = \hat{a}^\dagger$. The first-order commutator evaluates to:

$$\begin{aligned}
[\hat{A}, \hat{a}^\dagger] &= [-\alpha\hat{a}^\dagger + \alpha^*\hat{a}, \hat{a}^\dagger] \\
&= -\alpha[\hat{a}^\dagger, \hat{a}^\dagger] + \alpha^*[\hat{a}, \hat{a}^\dagger] \\
&= 0 + \alpha^*(1) \\
&= \alpha^*
\end{aligned}$$

Once again, the resulting c-number α^* causes all higher-order nested commutators ($[\hat{A}, \alpha^*]$) to vanish. The expansion truncates cleanly, yielding:

$$\hat{D}^\dagger(\alpha)\hat{a}^\dagger\hat{D}(\alpha) = e^{-\alpha\hat{a}^\dagger + \alpha^*\hat{a}}\hat{a}^\dagger e^{\alpha\hat{a}^\dagger - \alpha^*\hat{a}} = \hat{a}^\dagger + \alpha^*$$

2.13 Squeeze Operator

2.13.1 The unitary squeeze operator

It is possible to generate a squeezed single-mode state from an unsqueezed one by the action of the following simple unitary operator,

$$\hat{S}(\xi) = \exp\frac{1}{2}(\xi^*\hat{a}^2 - \xi\hat{a}^{\dagger 2}), \quad \xi = re^{i\theta},$$

which is known as the squeeze operator.

To find the mode evolution of the annihilation operator $\hat{a}$ under the single-mode

squeezing operator $\hat{S}(\xi)$, we apply the Baker-Campbell-Hausdorff (BCH) Lemma.

2.13.2 Defining the Operators

The squeezing operator for a single boson is defined using the squeezing parameter ξ:

$$\hat{S}(\xi) = \exp\left(\frac{1}{2}\xi^*\hat{a}^2 - \frac{1}{2}\xi(\hat{a}^\dagger)^2\right) \tag{101}$$

This squeezing parameter ξ is complex and can be expressed as $\xi = re^{i\theta}$, a value determined by the squeezing amplitude r and the squeezing angle θ.

Let the exponent be the operator $\hat{G}$, meaning $\hat{S}(\xi) = \exp\left(\hat{G}\right)$:

$$\hat{G} = \frac{1}{2}\xi^*\hat{a}^2 - \frac{1}{2}\xi(\hat{a}^\dagger)^2$$

Because $\hat{G}$ is an anti-Hermitian operator ($\hat{G}^\dagger = -\hat{G}$), the adjoint of the squeezing operator is $\hat{S}(\xi)^\dagger = \exp\left(-\hat{G}\right)$. The full transformation we need to calculate is therefore:

$$\hat{S}(\xi)^\dagger\hat{a}\hat{S}(\xi) = \exp\left(-\hat{G}\right)\hat{a}\exp\left(\hat{G}\right)$$

2.13.3 Applying the Baker-Campbell-Hausdorff Lemma

For generic operators $\hat{A}$ and $\hat{B}$, the BCH lemma states:

$$e^{\hat{A}}\hat{B}e^{-\hat{A}} = \hat{B} + [\hat{A}, \hat{B}] + \frac{1}{2!}[\hat{A}, [\hat{A}, \hat{B}]] + \frac{1}{3!}[\hat{A}, [\hat{A}, [\hat{A}, \hat{B}]]] + \dots$$

Substituting $\hat{A} = -\hat{G}$ and $\hat{B} = \hat{a}$, the formula generates an alternating series of nested commutators:

$$e^{-\hat{G}}\hat{a}e^{\hat{G}} = \hat{a} - [\hat{G}, \hat{a}] + \frac{1}{2!}[\hat{G}, [\hat{G}, \hat{a}]] - \frac{1}{3!}[\hat{G}, [\hat{G}, [\hat{G}, \hat{a}]]] + \dots$$

2.13.4 Calculating the Nested Commutators

We systematically calculate these individual commutators using the canonical commutation relation $[\hat{a}, \hat{a}^\dagger] = 1$.

First Commutator:

$$\begin{aligned}[\hat{G}, \hat{a}] &= \left[\frac{1}{2}\xi^*\hat{a}^2 - \frac{1}{2}\xi(\hat{a}^\dagger)^2, \hat{a}\right] \\ &= -\frac{1}{2}\xi[(\hat{a}^\dagger)^2, \hat{a}] + \frac{1}{2}\xi^*[\hat{a}^2, \hat{a}]\end{aligned}$$

Because any operator commutes with itself, $[\hat{a}^2, \hat{a}] = 0$. Using the identity $[\hat{A}\hat{B}, \hat{C}] = \hat{A}[\hat{B}, \hat{C}] + [\hat{A}, \hat{C}]\hat{B}$, we expand the remaining term:

$$\begin{aligned}[(\hat{a}^\dagger)^2, \hat{a}] &= \hat{a}^\dagger[\hat{a}^\dagger, \hat{a}] + [\hat{a}^\dagger, \hat{a}]\hat{a}^\dagger \\ &= \hat{a}^\dagger(-1) + (-1)\hat{a}^\dagger = -2\hat{a}^\dagger\end{aligned}$$

Substituting this result back yields:

$$[\hat{G}, \hat{a}] = -\frac{1}{2}\xi(-2\hat{a}^\dagger) = \xi\hat{a}^\dagger$$

Second Commutator:

$$\begin{aligned}[\hat{G}, [\hat{G}, \hat{a}]] &= [\hat{G}, \xi\hat{a}^\dagger] = \xi[\hat{G}, \hat{a}^\dagger] \\ [\hat{G}, \hat{a}^\dagger] &= \left[\frac{1}{2}\xi^*\hat{a}^2 - \frac{1}{2}\xi(\hat{a}^\dagger)^2, \hat{a}^\dagger\right] = \frac{1}{2}\xi^*[\hat{a}^2, \hat{a}^\dagger]\end{aligned}$$

Expanding $[\hat{a}^2, \hat{a}^\dagger]$:

$$[\hat{a}^2, \hat{a}^\dagger] = \hat{a}[\hat{a}, \hat{a}^\dagger] + [\hat{a}, \hat{a}^\dagger]\hat{a} = \hat{a}(1) + (1)\hat{a} = 2\hat{a}$$

Evaluating the full bracket gives:

$$[\hat{G}, \hat{a}^\dagger] = \frac{1}{2}\xi^*(2\hat{a}) = \xi^*\hat{a}$$

Returning to the nested commutator:

$$[\hat{G}, [\hat{G}, \hat{a}]] = \xi(\xi^*\hat{a}) = |\xi|^2\hat{a} = r^2\hat{a}$$

Higher-Order Commutators: The pattern predictably scales by r^2 with each cycle:

$$\begin{aligned}[\hat{G}, [\hat{G}, [\hat{G}, \hat{a}]]] &= [\hat{G}, r^2\hat{a}] = r^2[\hat{G}, \hat{a}] = r^2\xi\hat{a}^\dagger \\ [\hat{G}, [\hat{G}, [\hat{G}, [\hat{G}, \hat{a}]]]] &= [\hat{G}, r^2\xi\hat{a}^\dagger] = r^2\xi[\hat{G}, \hat{a}^\dagger] = r^4\hat{a}\end{aligned}$$

2.13.5 Summing the BCH Series

Substituting these evaluated commutators back into the original expansion:

$$\hat{S}^\dagger(\xi)\hat{a}\hat{S}(\xi) = \hat{a} - (\xi\hat{a}^\dagger) + \frac{1}{2!}(r^2\hat{a}) - \frac{1}{3!}(r^2\xi\hat{a}^\dagger) + \frac{1}{4!}(r^4\hat{a}) - \dots$$

Collecting terms containing $\hat{a}$ and $\hat{a}^\dagger$:

$$\hat{S}^\dagger(\xi)\hat{a}\hat{S}(\xi) = \hat{a}\left(1 + \frac{r^2}{2!} + \frac{r^4}{4!} + \dots\right) - \xi\hat{a}^\dagger\left(1 + \frac{r^2}{3!} + \frac{r^4}{5!} + \dots\right)$$

Substituting $\xi = re^{i\theta}$ and factoring out r for the second series:

$$\hat{S}^\dagger(\xi)\hat{a}\hat{S}(\xi) = \hat{a}\left(1 + \frac{r^2}{2!} + \frac{r^4}{4!} + \dots\right) - e^{i\theta}\hat{a}^\dagger\left(r + \frac{r^3}{3!} + \frac{r^5}{5!} + \dots\right)$$

These are the Taylor series for $\cosh(r)$ and $\sinh(r)$:

$$\hat{S}^\dagger(\xi)\hat{a}\hat{S}(\xi) = \cosh(r)\hat{a} - e^{i\theta}\sinh(r)\hat{a}^\dagger$$

2.13.6 Final Expression

Defining $\mu = \cosh r$ and $\nu = e^{i\theta}\sinh r$:

$$\hat{S}(\xi)^\dagger\hat{a}\hat{S}(\xi) = \mu\hat{a} - \nu\hat{a}^\dagger \tag{102}$$

2.14 Derivation of the Normal-Ordered Squeezing Operator using Operator Algebra

Our goal is to rewrite the single-mode squeezing operator $\hat{S}(\xi)$ into normal order, where all creation operators ($\hat{a}^\dagger$) are on the left, the number operators ($\hat{n} = \hat{a}^\dagger\hat{a}$) are in the middle, and the annihilation operators ($\hat{a}$) are on the right.

2.14.1 Constructing the Normal-Ordered Ansatz

Let us propose a general normal-ordered ansatz with unknown scalar coefficients A, B, and C, and a normalization constant M:

$$\hat{S} = Me^{A(\hat{a}^\dagger)^2}e^{B\hat{a}^\dagger\hat{a}}e^{C\hat{a}^2}$$

Because the squeezing operator is unitary, its inverse is its adjoint ($\hat{S}^{-1} = \hat{S}^\dagger$). From the work above, we know how this operator transforms the fundamental mode operators:

$$\hat{S}^{-1}\hat{a}\hat{S} = \mu\hat{a} - \nu\hat{a}^\dagger$$
$$\hat{S}^{-1}\hat{a}^\dagger\hat{S} = \mu\hat{a}^\dagger - \nu^*\hat{a}$$

where $\mu = \cosh r$ and $\nu = e^{i\theta}\sinh r$.

If we push the operators $\hat{a}$ and $\hat{a}^\dagger$ through our ansatz using the Baker-Campbell-Hausdorff (BCH) lemma and equate the results to the known transformations, we can solve for A, B, and C.

2.14.2 Pushing the Creation Operator Through the Ansatz

We evaluate the transformation of $\hat{a}^\dagger$ using our proposed form:

$$\hat{S}^{-1}\hat{a}^\dagger\hat{S} = \left(e^{-C\hat{a}^2}e^{-B\hat{a}^\dagger\hat{a}}e^{-A(\hat{a}^\dagger)^2}M^{-1}\right)\hat{a}^\dagger\left(Me^{A(\hat{a}^\dagger)^2}e^{B\hat{a}^\dagger\hat{a}}e^{C\hat{a}^2}\right)$$

The scalars M and M^{-1} cancel out. We move $\hat{a}^\dagger$ from the center to the right using the BCH expansion: $e^{\hat{X}}\hat{Y}e^{-\hat{X}} = \hat{Y} + [\hat{X},\hat{Y}] + \frac{1}{2!}[\hat{X},[\hat{X},\hat{Y}]] + \dots$

Step 2a: Commute through the A term

We evaluate $e^{-A(\hat{a}^\dagger)^2}\hat{a}^\dagger e^{A(\hat{a}^\dagger)^2}$. Because $[\hat{a}^\dagger,\hat{a}^\dagger] = 0$, all commutators are zero. The operator passes through unchanged:

$$\hat{a}^\dagger$$

Step 2b: Commute through the B term

We evaluate $e^{-B\hat{n}}\hat{a}^\dagger e^{B\hat{n}}$. Using $\hat{X} = -B\hat{n}$ and $\hat{Y} = \hat{a}^\dagger$, and knowing $[\hat{n},\hat{a}^\dagger] = \hat{a}^\dagger$:

$$[-B\hat{n},\hat{a}^\dagger] = -B\hat{a}^\dagger$$
$$[-B\hat{n},-B\hat{a}^\dagger] = B^2\hat{a}^\dagger$$

Summing the series gives:

$$\hat{a}^\dagger\left(1 - B + \frac{B^2}{2!} - \dots\right) = e^{-B}\hat{a}^\dagger$$

Step 2c: Commute through the C term

We evaluate $e^{-C\hat{a}^2}(e^{-B}\hat{a}^\dagger)e^{C\hat{a}^2}$. Using $\hat{X} = -C\hat{a}^2$ and $\hat{Y} = \hat{a}^\dagger$, and knowing $[\hat{a}^2,\hat{a}^\dagger] = 2\hat{a}$:

$$[-C\hat{a}^2,\hat{a}^\dagger] = -C(2\hat{a}) = -2C\hat{a}$$
$$[-C\hat{a}^2,-2C\hat{a}] = 2C^2[\hat{a}^2,\hat{a}] = 0$$

The series truncates immediately, yielding:

$$e^{-B}(\hat{a}^\dagger - 2C\hat{a})$$

Solving for B and C:

We equate our final expression to the known transformation:

$$e^{-B}\hat{a}^\dagger - 2Ce^{-B}\hat{a} = \mu\hat{a}^\dagger - \nu^*\hat{a}$$

Matching the coefficients of $\hat{a}^\dagger$:

$$e^{-B} = \mu \implies B = -\ln\mu$$

Matching the coefficients of $\hat{a}$:

$$-2C(e^{-B}) = -\nu^* \implies -2C\mu = -\nu^* \implies C = \frac{\nu^*}{2\mu}$$

2.14.3 Pushing the Annihilation Operator Through the Ansatz

Now we evaluate the transformation of $\hat{a}$ to find A:

$$\hat{S}^{-1}\hat{a}\hat{S} = e^{-C\hat{a}^2}e^{-B\hat{a}^\dagger\hat{a}}e^{-A(\hat{a}^\dagger)^2}\hat{a}e^{A(\hat{a}^\dagger)^2}e^{B\hat{a}^\dagger\hat{a}}e^{C\hat{a}^2}$$

Step 3a: Commute through the A term
Evaluate $e^{-A(\hat{a}^\dagger)^2}\hat{a}e^{A(\hat{a}^\dagger)^2}$. Knowing $[(\hat{a}^\dagger)^2, \hat{a}] = -2\hat{a}^\dagger$:

$$\hat{a} + 2A\hat{a}^\dagger$$

Step 3b: Commute through the B term
Evaluate $e^{-B\hat{n}}(\hat{a} + 2A\hat{a}^\dagger)e^{B\hat{n}}$. We apply the linear transformation to both terms. From Step 2b, $\hat{a}^\dagger \to e^{-B}\hat{a}^\dagger$. For the $\hat{a}$ term, knowing $[\hat{n}, \hat{a}] = -\hat{a}$, the BCH series evaluates to $e^B\hat{a}$. The result is:

$$e^B\hat{a} + 2Ae^{-B}\hat{a}^\dagger$$

Step 3c: Commute through the C term
Evaluate $e^{-C\hat{a}^2}(e^B\hat{a} + 2Ae^{-B}\hat{a}^\dagger)e^{C\hat{a}^2}$. Since $[\hat{a}^2, \hat{a}] = 0$, the first term is untouched. From Step 2c, $\hat{a}^\dagger \to \hat{a}^\dagger - 2C\hat{a}$.

$$e^B\hat{a} + 2Ae^{-B}(\hat{a}^\dagger - 2C\hat{a}) = (e^B - 4ACe^{-B})\hat{a} + 2Ae^{-B}\hat{a}^\dagger$$

Solving for A:
Equate the coefficient of $\hat{a}^\dagger$ to the known transformation $(\mu\hat{a} - \nu\hat{a}^\dagger)$:

$$2Ae^{-B} = -\nu \implies 2A\mu = -\nu \implies A = -\frac{\nu}{2\mu}$$

2.14.4 Determining the Normalization Scalar

Our ansatz is now populated with the operational coefficients:

$$\hat{S} = M\exp\left(-\frac{\nu}{2\mu}(\hat{a}^\dagger)^2\right)\exp\left(-(\ln\mu)\hat{a}^\dagger\hat{a}\right)\exp\left(\frac{\nu^*}{2\mu}\hat{a}^2\right)$$

We apply this operator to the vacuum state $|0\rangle$ to enforce the norm condition $\langle\xi|\xi\rangle = 1$, where $|\xi\rangle = \hat{S}|0\rangle$. Because $\hat{a}^2|0\rangle = 0$ and $\hat{a}^\dagger\hat{a}|0\rangle = 0$, the two rightmost exponentials act

as the identity:

$$|\xi\rangle = M \exp\left(-\frac{\nu}{2\mu}(\hat{a}^\dagger)^2\right)|0\rangle$$

Expanding this into a Taylor series and applying it to the vacuum yields the Fock state expansion:

$$|\xi\rangle = M \sum_{k=0}^{\infty} \frac{1}{k!}\left(-\frac{\nu}{2\mu}\right)^k \sqrt{(2k)!}|2k\rangle$$

Squaring this to find the total probability (the norm), which must equal 1:

$$\langle\xi|\xi\rangle = |M|^2 \sum_{k=0}^{\infty} \frac{(2k)!}{(k!)^2}\left|\frac{\nu}{2\mu}\right|^{2k} = 1$$

Using the binomial series for negative fractional powers, $(1-x)^{-1/2} = \sum_{k=0}^{\infty} \frac{(2k)!}{(k!)^2}\left(\frac{x}{4}\right)^k$, we substitute $x = \tanh^2 r$ (since $\frac{\tanh^2 r}{4} = \left|\frac{\nu}{2\mu}\right|^2$).

$$\sum_{k=0}^{\infty} \frac{(2k)!}{(k!)^2}\left|\frac{\nu}{2\mu}\right|^{2k} = (1-\tanh^2 r)^{-1/2} = (\mathrm{sech}^2 r)^{-1/2} = \cosh r = \mu$$

Substituting this back into the norm equation:

$$|M|^2\mu = 1 \implies M = \mu^{-1/2}$$

2.14.5 Final Assembly

We place M back into the full operational expression:

$$\hat{S}(\xi) = \mu^{-1/2} \exp\left(-\frac{\nu}{2\mu}(\hat{a}^\dagger)^2\right) \exp\left(-(\ln\mu)\hat{a}^\dagger\hat{a}\right) \exp\left(\frac{\nu^*}{2\mu}\hat{a}^2\right)$$

We combine the scalar $\mu^{-1/2}$ into the middle exponent using logarithm rules ($\mu^y = e^{y\ln\mu}$):

$$\mu^{-1/2} \exp\left(-(\ln\mu)\hat{a}^\dagger\hat{a}\right) = \mu^{-1/2}\mu^{-\hat{a}^\dagger\hat{a}} = \mu^{-(\hat{a}^\dagger\hat{a}+1/2)}$$

Yielding the final, exact normal-ordered form of the squeezing operator:

$$\hat{S}(\xi) = \exp\left(-\frac{\nu}{2\mu}(\hat{a}^\dagger)^2\right) \mu^{-(\hat{a}^\dagger\hat{a}+\frac{1}{2})} \exp\left(\frac{\nu^*}{2\mu}\hat{a}^2\right) \tag{103}$$

2.14.6 Squeezed state form

We have stated earlier:

$$|\xi\rangle = M \sum_{k=0}^{\infty} \frac{1}{k!} \left(-\frac{\nu}{2\mu}\right)^k \sqrt{(2k)!}|2k\rangle$$

Thus, we have

$$|\xi\rangle = \mu^{-1/2} \sum_{k=0}^{\infty} \frac{1}{k!} \left(-\frac{\nu}{2\mu}\right)^k \sqrt{(2k)!}|2k\rangle \tag{104}$$

which is also a consequent result of our ansatz based approach.

2.15 Derivation of the Wigner Function for the Squeezed Vacuum State

This derivation approaches the Wigner function $W(x, p)$ of the squeezed vacuum state by first determining its position wavefunction, and then applying the Fourier transform definition of the Wigner function.

2.15.1 The Annihilation Condition

For a state created by acting on the vacuum $|0\rangle$ with a unitary operator, the resulting state is the "vacuum" for the correspondingly transformed annihilation operator.

For the squeezed vacuum state $|\xi\rangle = \hat{S}(\xi)|0\rangle$, we define a transformed annihilation operator $\hat{b}$:

$$\hat{b} = \hat{S}(\xi)\hat{a}\hat{S}^\dagger(\xi)$$

Because the standard annihilation operator destroys the vacuum ($\hat{a}|0\rangle = 0$), the transformed operator must destroy the squeezed vacuum:

$$\hat{b}|\xi\rangle = \hat{S}(\xi)\hat{a}\hat{S}^\dagger(\xi)\hat{S}(\xi)|0\rangle = \hat{S}(\xi)\hat{a}|0\rangle = 0$$

2.15.2 Formulating $\hat{b}$ in Phase Space

From the known mode evolution, $\hat{S}^\dagger(\xi)\hat{a}\hat{S}(\xi) = \mu\hat{a} - \nu\hat{a}^\dagger$. To find $\hat{S}(\xi)\hat{a}\hat{S}^\dagger(\xi)$, we invert the operation by replacing the squeezing parameter ξ with $-\xi$, which flips the sign of ν:

$$\hat{b} = \mu\hat{a} + \nu\hat{a}^\dagger$$

Assuming a purely real squeezing parameter $\xi = r$ (setting the angle $\theta = 0$), we have $\mu = \cosh r$ and $\nu = \sinh r$:

$$\hat{b} = \cosh(r)\hat{a} + \sinh(r)\hat{a}^\dagger$$

We express the creation and annihilation operators in terms of the position and momentum quadratures, $\hat{a} = \hat{q} + i\hat{p}$ and $\hat{a}^\dagger = \hat{q} - i\hat{p}$:

$$\begin{aligned}\hat{b} &= \cosh(r)(\hat{q} + i\hat{p}) + \sinh(r)(\hat{q} - i\hat{p}) \\ &= (\cosh r + \sinh r)\hat{q} + i(\cosh r - \sinh r)\hat{p}\end{aligned}$$

Using the identities $e^{\pm r} = \cosh r \pm \sinh r$, we finalize the operator:

$$\hat{b} = e^r\hat{q} + ie^{-r}\hat{p}$$

2.15.3 Solving the Position Wavefunction

We apply the condition $\hat{b}|\xi\rangle = 0$ in the continuous position basis to solve for the wavefunction $\psi(x) = \langle x|\xi\rangle$.

The commutation relation $[\hat{q}, \hat{p}] = i/2$ dictates the representations $\hat{q} \to x$ and $\hat{p} \to -\frac{i}{2}\frac{d}{dx}$. Substituting these yields a first-order differential equation:

$$\begin{aligned}
\left(e^r x + ie^{-r}\left(-\frac{i}{2}\frac{d}{dx}\right)\right)\psi(x) &= 0 \\
e^r x\psi(x) + \frac{1}{2}e^{-r}\frac{d\psi(x)}{dx} &= 0 \\
\frac{d\psi(x)}{dx} &= -2e^{2r}x\psi(x)
\end{aligned}$$

Integrating this separable differential equation produces a Gaussian distribution with a normalization constant N:

$$\begin{aligned}
\ln\psi(x) &= -e^{2r}x^2 + C \\
\psi(x) &= N\exp\left(-e^{2r}x^2\right)
\end{aligned}$$

We find N by enforcing the probability normalization $\int_{-\infty}^{\infty}|\psi(x)|^2 dx = 1$:

$$|N|^2\int_{-\infty}^{\infty}\exp\left(-2e^{2r}x^2\right)dx = 1$$

Using the standard Gaussian integral $\int e^{-ax^2}dx = \sqrt{\pi/a}$ with $a = 2e^{2r}$:

$$|N|^2\sqrt{\frac{\pi}{2e^{2r}}} = 1 \implies N = \left(\frac{2e^{2r}}{\pi}\right)^{1/4}$$

The exact, normalized position wavefunction is:

$$\psi(x) = \left(\frac{2e^{2r}}{\pi}\right)^{1/4}\exp\left(-e^{2r}x^2\right)$$

2.15.4 Integrating the Wigner Function

The Wigner function for a pure state is:

$$W(x,p) = \frac{1}{\pi\hbar}\int_{-\infty}^{\infty}\psi^*(x+y)\psi(x-y)e^{2ipy/\hbar}dy$$

Because $[\hat{q}, \hat{p}] = i/2$, the effective value for $\hbar$ is $1/2$. This simplifies the expression to:

$$W(x,p) = \frac{2}{\pi}\int_{-\infty}^{\infty}\psi^*(x+y)\psi(x-y)e^{4ipy}dy$$

We explicitly evaluate the product of the wavefunctions:

$$\begin{aligned}\psi^*(x+y)\psi(x-y) &= \sqrt{\frac{2e^{2r}}{\pi}} \exp\left[-e^{2r}(x+y)^2 - e^{2r}(x-y)^2\right] \\ &= \sqrt{\frac{2e^{2r}}{\pi}} \exp\left(-2e^{2r}x^2 - 2e^{2r}y^2\right)\end{aligned}$$

Substituting this back into the Wigner integral and pulling independent terms outside:

$$W(x,p) = \frac{2}{\pi}\sqrt{\frac{2e^{2r}}{\pi}} \exp\left(-2e^{2r}x^2\right) \int_{-\infty}^{\infty} \exp\left(-2e^{2r}y^2\right) e^{4ipy} dy$$

The integral is a standard Fourier transform of a Gaussian, $\int e^{-ay^2+by}dy = \sqrt{\frac{\pi}{a}}e^{b^2/4a}$, with $a = 2e^{2r}$ and $b = 4ip$:

$$\frac{b^2}{4a} = \frac{(4ip)^2}{4(2e^{2r})} = \frac{-16p^2}{8e^{2r}} = -2e^{-2r}p^2$$

The integral evaluates to:

$$\sqrt{\frac{\pi}{2e^{2r}}} \exp\left(-2e^{-2r}p^2\right)$$

Multiplying this by the prefactors:

$$W(x,p) = \frac{2}{\pi}\sqrt{\frac{2e^{2r}}{\pi}} \exp\left(-2e^{2r}x^2\right) \sqrt{\frac{\pi}{2e^{2r}}} \exp\left(-2e^{-2r}p^2\right)$$

The square root multipliers cancel out exactly ($\sqrt{A/\pi} \cdot \sqrt{\pi/A} = 1$). The final Wigner function is:

$$W(x,p) = \frac{2}{\pi} \exp\left(-2e^{2r}x^2 - 2e^{-2r}p^2\right) \tag{105}$$

2.16 Algebraic Derivation of Two-Mode Squeezing

2.16.1 Definitions

We would now like to move from the single-mode formalism to two independent bosonic modes, which is required for the generation of continuous-variable entanglement. Let $\hat{a}$ and $\hat{b}$ represent independent modes.

$$[\hat{a}, \hat{a}^\dagger] = 1, \quad [\hat{b}, \hat{b}^\dagger] = 1$$
$$[\hat{a}, \hat{b}] = [\hat{a}, \hat{b}^\dagger] = [\hat{a}^\dagger, \hat{b}] = 0$$

The two-mode squeezing operator is defined as:

$$\hat{S}_2(\xi) = \exp\Big(\xi \hat{a}^\dagger \hat{b}^\dagger - \xi^* \hat{a}\hat{b}\Big)$$

where $\xi = re^{i\theta}$.

2.16.2 Mode Evolution

We evaluate $\hat{S}_2^\dagger(\xi)\hat{a}\hat{S}_2(\xi)$. Define the exponent $\hat{K} = \xi^*\hat{a}\hat{b} - \xi\hat{a}^\dagger\hat{b}^\dagger$. Using the Baker-Campbell-Hausdorff (BCH) (similar to the one for single mode):

$$e^{\hat{K}}\hat{a}e^{-\hat{K}} = \hat{a} + [\hat{K}, \hat{a}] + \frac{1}{2!}[\hat{K}, [\hat{K}, \hat{a}]] + \dots$$

Evaluating the commutators:

$$[\hat{K}, \hat{a}] = [\xi^*\hat{a}\hat{b} - \xi\hat{a}^\dagger\hat{b}^\dagger, \hat{a}] = [-\xi\hat{a}^\dagger\hat{b}^\dagger, \hat{a}] = -\xi[\hat{a}^\dagger, \hat{a}]\hat{b}^\dagger = \xi\hat{b}^\dagger$$
$$[\hat{K}, [\hat{K}, \hat{a}]] = [\xi^*\hat{a}\hat{b} - \xi\hat{a}^\dagger\hat{b}^\dagger, \xi\hat{b}^\dagger] = [\xi^*\hat{a}\hat{b}, \xi\hat{b}^\dagger] = \xi\xi^*\hat{a}[\hat{b}, \hat{b}^\dagger] = r^2\hat{a}$$

Substituting back into the series:

$$\hat{S}_2^\dagger\hat{a}\hat{S}_2 = \hat{a}\left(1 + \frac{r^2}{2!} + \frac{r^4}{4!} + \dots\right) + \xi\hat{b}^\dagger\left(1 + \frac{r^2}{3!} + \frac{r^4}{5!} + \dots\right)$$

Substituting $\xi = e^{i\theta}r$, $\mu = \cosh r$, and $\nu = e^{i\theta}\sinh r$:

$$\hat{S}_2^\dagger\hat{a}\hat{S}_2 = \mu\hat{a} + \nu\hat{b}^\dagger \tag{106}$$

By symmetry for mode $\hat{b}$:

$$\hat{S}_2^\dagger\hat{b}^\dagger\hat{S}_2 = \nu^*\hat{a} + \mu\hat{b}^\dagger \tag{107}$$

2.16.3 Normal-Ordered Ladder Expansion

Propose an algebraic ansatz with coefficients A, B, C and scalar M:

$$\hat{S}_2 = M e^{A\hat{a}^\dagger \hat{b}^\dagger} e^{B(\hat{a}^\dagger \hat{a} + \hat{b}^\dagger \hat{b})} e^{C\hat{a}\hat{b}}$$

The inverse is:

$$\hat{S}_2^{-1} = e^{-C\hat{a}\hat{b}} e^{-B(\hat{n}_a + \hat{n}_b)} e^{-A\hat{a}^\dagger \hat{b}^\dagger} M^{-1}$$

Step 3a: Push $\hat{b}^\dagger$ through $\hat{S}_2^{-1}$

$$\begin{aligned}
&\text{Through A:} \quad e^{-A\hat{a}^\dagger \hat{b}^\dagger} \hat{b}^\dagger e^{A\hat{a}^\dagger \hat{b}^\dagger} = \hat{b}^\dagger \\
&\text{Through B:} \quad e^{-B\hat{n}_b} \hat{b}^\dagger e^{B\hat{n}_b} \implies [-B\hat{n}_b, \hat{b}^\dagger] = -B\hat{b}^\dagger \implies e^{-B}\hat{b}^\dagger \\
&\text{Through C:} \quad e^{-C\hat{a}\hat{b}} (e^{-B}\hat{b}^\dagger) e^{C\hat{a}\hat{b}} \implies [-C\hat{a}\hat{b}, \hat{b}^\dagger] = -C\hat{a} \implies e^{-B}(\hat{b}^\dagger - C\hat{a})
\end{aligned}$$

Equating to the known evolution $\hat{S}_2^{-1}\hat{b}^\dagger \hat{S}_2 = \nu^* \hat{a} + \mu \hat{b}^\dagger$:

$$e^{-B}\hat{b}^\dagger - Ce^{-B}\hat{a} = \mu\hat{b}^\dagger + \nu^*\hat{a} \implies B = -\ln\mu, \quad C = -\frac{\nu^*}{\mu}$$

Step 3b: Push $\hat{a}$ through $\hat{S}_2^{-1}$

$$\begin{aligned}
&\text{Through A:} \quad e^{-A\hat{a}^\dagger \hat{b}^\dagger} \hat{a} e^{A\hat{a}^\dagger \hat{b}^\dagger} \implies [-A\hat{a}^\dagger \hat{b}^\dagger, \hat{a}] = A\hat{b}^\dagger \implies \hat{a} + A\hat{b}^\dagger \\
&\text{Through B:} \quad e^{-B(\hat{n}_a + \hat{n}_b)} (\hat{a} + A\hat{b}^\dagger) e^{B(\hat{n}_a + \hat{n}_b)} \implies e^B \hat{a} + Ae^{-B}\hat{b}^\dagger \\
&\text{Through C:} \quad e^{-C\hat{a}\hat{b}} (e^B \hat{a} + Ae^{-B}\hat{b}^\dagger) e^{C\hat{a}\hat{b}} \implies e^B \hat{a} + Ae^{-B}(\hat{b}^\dagger - C\hat{a}) \\
&\qquad\qquad\qquad\qquad = (e^B - ACe^{-B})\hat{a} + Ae^{-B}\hat{b}^\dagger
\end{aligned}$$

Equating to $\mu\hat{a} + \nu\hat{b}^\dagger$:

$$Ae^{-B} = \nu \implies A\mu = \nu \implies A = \frac{\nu}{\mu}$$

Step 3c: Determining Normalization M Applying the populated ansatz to the vacuum $|0,0\rangle$, we have:

$$|EPR\rangle = \hat{S}_2|0,0\rangle = M \exp\left(\frac{\nu}{\mu}\hat{a}^\dagger \hat{b}^\dagger\right)|0,0\rangle = M \sum_{k=0}^{\infty} \left(\frac{\nu}{\mu}\right)^k |k,k\rangle$$

Enforce normalization $\langle\psi|\psi\rangle = 1$:

$$
\begin{aligned}
|M|^2 \sum_{k=0}^{\infty} \left|\frac{\nu}{\mu}\right|^{2k} &= |M|^2 \sum_{k=0}^{\infty} (\tanh^2 r)^k \\
&= |M|^2 \left(\frac{1}{1-\tanh^2 r}\right) = |M|^2 \cosh^2 r = 1 \implies M = \mu^{-1}
\end{aligned}
$$

Step 3d: Final Assembly Substitute M and combine the middle exponent terms using $\mu^y = e^{y \ln \mu}$:

$$
\hat{S}_2(\xi) = \mu^{-1} \exp\left(\frac{\nu}{\mu}\hat{a}^\dagger \hat{b}^\dagger\right) \exp\left(-(\ln \mu)(\hat{a}^\dagger \hat{a} + \hat{b}^\dagger \hat{b})\right) \exp\left(-\frac{\nu^*}{\mu}\hat{a}\hat{b}\right) \tag{108}
$$

$$
\hat{S}_2(\xi) = \exp\left(\frac{\nu}{\mu}\hat{a}^\dagger \hat{b}^\dagger\right) \mu^{-(\hat{a}^\dagger \hat{a} + \hat{b}^\dagger \hat{b} + 1)} \exp\left(-\frac{\nu^*}{\mu}\hat{a}\hat{b}\right) \tag{109}
$$

2.16.4 Sidenote :

From the normalization step for M, we can directly substitute and arrive at the state eqn. :

$$
|EPR\rangle = \mu^{-1} \sum_{k=0}^{\infty} \left(\frac{\nu}{\mu}\right)^k |k, k\rangle \tag{110}
$$

2.17 Creating Gaussian EPR states

2.17.1 Preliminary Properties for Usage

Our aim here is to evaluate the expectation values of the mode operators for the squeezed vacuum state $|\xi\rangle = \hat{S}(\xi)|0\rangle$. We shall be relying on the known evolutions: $\hat{S}^\dagger \hat{a} \hat{S} = \mu\hat{a} - \nu\hat{a}^\dagger$ and $\hat{S}^\dagger \hat{a}^\dagger \hat{S} = \mu\hat{a}^\dagger - \nu^*\hat{a}$, where $\mu = \cosh r$ and $\nu = e^{i\theta}\sinh r$.

The core rule for these derivations is that the annihilation operator destroys the vacuum ($\hat{a}|0\rangle = 0$), and its dual creates zero when acting to the left ($\langle 0|\hat{a}^\dagger = 0$).

2.17.2 The First Moments (Means)

We begin by proving that the expectation values of the fundamental mode operators are strictly zero.

$$\begin{aligned}\langle\xi|\hat{a}|\xi\rangle &= \langle 0|\hat{S}^\dagger \hat{a} \hat{S}|0\rangle \\ &= \langle 0|(\mu\hat{a} - \nu\hat{a}^\dagger)|0\rangle \\ &= \mu\langle 0|\hat{a}|0\rangle - \nu\langle 0|\hat{a}^\dagger|0\rangle\end{aligned}$$

Because $\hat{a}|0\rangle = 0$ and $\langle 0|\hat{a}^\dagger = 0$, both terms independently evaluate to zero:

$$\langle\xi|\hat{a}|\xi\rangle = 0$$

By applying the exact same logic and taking the complex conjugate, the creation operator mean is also zero:

$$\langle\xi|\hat{a}^\dagger|\xi\rangle = 0$$

2.17.3 The Photon Number

Next, we prove that the expected photon number $\langle\hat{n}\rangle = \langle\hat{a}^\dagger\hat{a}\rangle$ equals $\sinh^2 r$. To do this, we insert the identity $\hat{I} = \hat{S}\hat{S}^\dagger$ between the operators.

$$\begin{aligned}\langle\xi|\hat{a}^\dagger\hat{a}|\xi\rangle &= \langle 0|\hat{S}^\dagger\hat{a}^\dagger(\hat{S}\hat{S}^\dagger)\hat{a}\hat{S}|0\rangle \\ &= \langle 0|(\hat{S}^\dagger\hat{a}^\dagger\hat{S})(\hat{S}^\dagger\hat{a}\hat{S})|0\rangle \\ &= \langle 0|(\mu\hat{a}^\dagger - \nu^*\hat{a})(\mu\hat{a} - \nu\hat{a}^\dagger)|0\rangle\end{aligned}$$

We explicitly expand this binomial product, ensuring no terms are skipped:

$$= \langle 0|\mu^2\hat{a}^\dagger\hat{a} - \mu\nu\hat{a}^\dagger\hat{a}^\dagger - \nu^*\mu\hat{a}\hat{a} + |\nu|^2\hat{a}\hat{a}^\dagger|0\rangle$$

We now evaluate the action of these four operators on the vacuum state $|0\rangle$:

1. $\hat{a}^\dagger\hat{a}|0\rangle = \hat{a}^\dagger(0) = 0$

2. $\hat{a}^\dagger\hat{a}^\dagger|0\rangle = \sqrt{2}|2\rangle$. Because $\langle 0|2\rangle = 0$, this term vanishes.

3. $\hat{a}\hat{a}|0\rangle = \hat{a}(0) = 0$

4. $\hat{a}\hat{a}^\dagger|0\rangle = \hat{a}|1\rangle = |0\rangle$. This is the only term that survives.

Substituting these evaluations back into our expansion yields:

$$\langle\xi|\hat{a}^\dagger\hat{a}|\xi\rangle = |\nu|^2\langle 0|0\rangle = |\nu|^2$$

Because $\nu = e^{i\theta}\sinh r$, its absolute square is $|\nu|^2 = \sinh^2 r$.

$$\langle\xi|\hat{a}^\dagger\hat{a}|\xi\rangle = \sinh^2 r \tag{111}$$

2.17.4 The Variance / Squeezing Terms

We now calculate the expectations of the paired annihilation and creation operators.

$$\begin{aligned}\langle\xi|\hat{a}\hat{a}|\xi\rangle &= \langle 0|(\hat{S}^\dagger\hat{a}\hat{S})(\hat{S}^\dagger\hat{a}\hat{S})|0\rangle \\ &= \langle 0|(\mu\hat{a} - \nu\hat{a}^\dagger)(\mu\hat{a} - \nu\hat{a}^\dagger)|0\rangle\end{aligned}$$

Expanding the binomial product completely:

$$= \langle 0|\mu^2\hat{a}\hat{a} - \mu\nu\hat{a}\hat{a}^\dagger - \nu\mu\hat{a}^\dagger\hat{a} + \nu^2\hat{a}^\dagger\hat{a}^\dagger|0\rangle$$

Using the exact same vacuum evaluation rules established above (from the annihilation and creation operators' action on vacuum state), only the $\hat{a}\hat{a}^\dagger$ operator survives acting on $|0\rangle$:

$$\langle\xi|\hat{a}\hat{a}|\xi\rangle = -\mu\nu\langle 0|\hat{a}\hat{a}^\dagger|0\rangle = -\mu\nu$$

Substituting our definitions for μ and ν:

$$\langle\xi|\hat{a}\hat{a}|\xi\rangle = -(\cosh r)(e^{i\theta}\sinh r) = -e^{i\theta}\sinh r\cosh r \tag{112}$$

By applying identical logic to the creation operators, we evaluate $\langle\hat{a}^\dagger\hat{a}^\dagger\rangle$:

$$\begin{aligned}\langle\xi|\hat{a}^\dagger\hat{a}^\dagger|\xi\rangle &= \langle 0|(\mu\hat{a}^\dagger - \nu^*\hat{a})(\mu\hat{a}^\dagger - \nu^*\hat{a})|0\rangle \\ &= \langle 0|\mu^2\hat{a}^\dagger\hat{a}^\dagger - \mu\nu^*\hat{a}^\dagger\hat{a} - \nu^*\mu\hat{a}\hat{a}^\dagger + (\nu^*)^2\hat{a}\hat{a}|0\rangle \\ &= -\mu\nu^*\langle 0|\hat{a}\hat{a}^\dagger|0\rangle = -\mu\nu^*\end{aligned}$$

Substituting the definitions yields the final statistic:

$$\langle\xi|\hat{a}^\dagger\hat{a}^\dagger|\xi\rangle = -e^{-i\theta}\sinh r\cosh r \tag{113}$$

2.18 Explicit Derivation of the Matrices for the Gaussian EPR State

The creation of a Gaussian EPR state involves passing two independent squeezed vacuum states through a balanced 50:50 beam-splitter. We construct the three fundamental matrices governing this evolution in phase space: the input covariance matrix (σ_{in}), the symplectic beam-splitter matrix (B), and the output covariance matrix (σ_{EPR}).

2.18.1 Deriving the Input Covariance Matrix

The continuous variable phase space is defined by the quadrature vector $\vec{R} = (\hat{q}_1, \hat{p}_1, \hat{q}_2, \hat{p}_2)^T$. The elements of the covariance matrix σ for zero-mean states are given by $\sigma_{ij} = \frac{1}{2}\langle \hat{R}_i \hat{R}_j + \hat{R}_j \hat{R}_i \rangle$.

We decompose the quadratures into annihilation and creation operators using $\hat{q} = \frac{1}{2}(\hat{a} + \hat{a}^\dagger)$ and $\hat{p} = \frac{1}{2i}(\hat{a} - \hat{a}^\dagger)$. For a single-mode squeezed state (assuming $\theta = 0$), the known expectations are:

$$\langle \hat{a}^\dagger \hat{a} \rangle = \sinh^2 r, \quad \langle \hat{a}\hat{a}^\dagger \rangle = \cosh^2 r, \quad \langle \hat{a}^2 \rangle = \langle (\hat{a}^\dagger)^2 \rangle = -\sinh r \cosh r$$

Evaluating Mode 1 (squeezed by parameter r):

$$\begin{aligned}
\langle \hat{q}_1^2 \rangle &= \frac{1}{4}\langle \hat{a}_1^2 + (\hat{a}_1^\dagger)^2 + \hat{a}_1\hat{a}_1^\dagger + \hat{a}_1^\dagger\hat{a}_1 \rangle \\
&= \frac{1}{4}\left(-2\sinh r \cosh r + \cosh^2 r + \sinh^2 r\right) \\
&= \frac{1}{4}(\cosh r - \sinh r)^2 = \frac{1}{4}e^{-2r}
\end{aligned}$$

$$\begin{aligned}
\langle \hat{p}_1^2 \rangle &= -\frac{1}{4}\langle \hat{a}_1^2 + (\hat{a}_1^\dagger)^2 - \hat{a}_1\hat{a}_1^\dagger - \hat{a}_1^\dagger\hat{a}_1 \rangle \\
&= -\frac{1}{4}\left(-2\sinh r \cosh r - \cosh^2 r - \sinh^2 r\right) \\
&= \frac{1}{4}(\cosh r + \sinh r)^2 = \frac{1}{4}e^{2r}
\end{aligned}$$

Evaluating Mode 2 (squeezed by parameter $-r$): Substituting $r \to -r$ into the above results directly flips the exponents:

$$\begin{aligned}
\langle \hat{q}_2^2 \rangle &= \frac{1}{4}e^{-2(-r)} = \frac{1}{4}e^{2r} \\
\langle \hat{p}_2^2 \rangle &= \frac{1}{4}e^{2(-r)} = \frac{1}{4}e^{-2r}
\end{aligned}$$

Because the input modes are independent, all cross-correlations between Mode 1 and Mode 2 are zero. The input state $\hat{S}(r)|0\rangle_1 \otimes \hat{S}(-r)|0\rangle_2$ is a product state, making its

covariance matrix a block-diagonal tensor sum of the two modes:

$$\sigma_{in} = \frac{1}{4}\begin{pmatrix} e^{-2r} & 0 & 0 & 0 \\ 0 & e^{2r} & 0 & 0 \\ 0 & 0 & e^{2r} & 0 \\ 0 & 0 & 0 & e^{-2r} \end{pmatrix}$$

2.18.2 Deriving the Symplectic Beam-Splitter Matrix

A standard beam-splitter transforms two input modes $(\hat{a}, \hat{b})$ into output modes $(\hat{c}, \hat{d})$. For a balanced 50:50 beam-splitter ($\eta = 1/2$), the canonical transformations are:

$$\hat{c} = \frac{1}{\sqrt{2}}\hat{a} + \frac{1}{\sqrt{2}}\hat{b}$$
$$\hat{d} = -\frac{1}{\sqrt{2}}\hat{a} + \frac{1}{\sqrt{2}}\hat{b}$$

Because the quadratures $\hat{q}$ and $\hat{p}$ are linear combinations of the mode operators, they inherit this exact same transformation geometry. Mapping the inputs $(\hat{q}_1, \hat{p}_1, \hat{q}_2, \hat{p}_2)$ to outputs $(\hat{q}_c, \hat{p}_c, \hat{q}_d, \hat{p}_d)$ produces a system of linear equations:

$$\hat{q}_c = \frac{1}{\sqrt{2}}\hat{q}_1 + \frac{1}{\sqrt{2}}\hat{q}_2$$
$$\hat{p}_c = \frac{1}{\sqrt{2}}\hat{p}_1 + \frac{1}{\sqrt{2}}\hat{p}_2$$
$$\hat{q}_d = -\frac{1}{\sqrt{2}}\hat{q}_1 + \frac{1}{\sqrt{2}}\hat{q}_2$$
$$\hat{p}_d = -\frac{1}{\sqrt{2}}\hat{p}_1 + \frac{1}{\sqrt{2}}\hat{p}_2$$

Expressing this linear transformation as a 4×4 matrix equation $\vec{R}_{out} = B\vec{R}_{in}$ yields the real symplectic matrix B:

$$B = \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} & 0 \\ 0 & -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{pmatrix}$$

2.18.3 Deriving the Output EPR Covariance Matrix

The final covariance matrix evolves under the symplectic transformation as $\sigma_{EPR} = B\sigma_{in}B^T$. We evaluate this via rigorous matrix multiplication.

Step 3a: Calculate the intermediate matrix $B\sigma_{in}$

$$B\sigma_{in} = \frac{1}{4}\begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} & 0 \\ 0 & -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{pmatrix}\begin{pmatrix} e^{-2r} & 0 & 0 & 0 \\ 0 & e^{2r} & 0 & 0 \\ 0 & 0 & e^{2r} & 0 \\ 0 & 0 & 0 & e^{-2r} \end{pmatrix}$$
$$= \frac{1}{4}\begin{pmatrix} \frac{e^{-2r}}{\sqrt{2}} & 0 & \frac{e^{2r}}{\sqrt{2}} & 0 \\ 0 & \frac{e^{2r}}{\sqrt{2}} & 0 & \frac{e^{-2r}}{\sqrt{2}} \\ -\frac{e^{-2r}}{\sqrt{2}} & 0 & \frac{e^{2r}}{\sqrt{2}} & 0 \\ 0 & -\frac{e^{2r}}{\sqrt{2}} & 0 & \frac{e^{-2r}}{\sqrt{2}} \end{pmatrix}$$

Step 3b: Calculate the final matrix $(B\sigma_{in})B^T$

$$(B\sigma_{in})B^T = \frac{1}{4}\begin{pmatrix} \frac{e^{-2r}}{\sqrt{2}} & 0 & \frac{e^{2r}}{\sqrt{2}} & 0 \\ 0 & \frac{e^{2r}}{\sqrt{2}} & 0 & \frac{e^{-2r}}{\sqrt{2}} \\ -\frac{e^{-2r}}{\sqrt{2}} & 0 & \frac{e^{2r}}{\sqrt{2}} & 0 \\ 0 & -\frac{e^{2r}}{\sqrt{2}} & 0 & \frac{e^{-2r}}{\sqrt{2}} \end{pmatrix}\begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{pmatrix}$$

We compute the dot products for the active non-zero elements:

- **Element (1,1) and (3,3):** $\frac{1}{4}\left(\frac{1}{2}e^{-2r} + \frac{1}{2}e^{2r}\right) = \frac{1}{4}\left(\frac{e^{2r}+e^{-2r}}{2}\right) = \frac{1}{4}\cosh 2r$
- **Element (2,2) and (4,4):** $\frac{1}{4}\left(\frac{1}{2}e^{2r} + \frac{1}{2}e^{-2r}\right) = \frac{1}{4}\cosh 2r$
- **Element (1,3) and (3,1):** $\frac{1}{4}\left(-\frac{1}{2}e^{-2r} + \frac{1}{2}e^{2r}\right) = \frac{1}{4}\left(\frac{e^{2r}-e^{-2r}}{2}\right) = \frac{1}{4}\sinh 2r$
- **Element (2,4) and (4,2):** $\frac{1}{4}\left(-\frac{1}{2}e^{2r} + \frac{1}{2}e^{-2r}\right) = \frac{1}{4}\left(\frac{e^{-2r}-e^{2r}}{2}\right) = -\frac{1}{4}\sinh 2r$

Populating the full array with these evaluated elements yields the exact output covariance matrix:

$$\sigma_{EPR} = \frac{1}{4}\begin{pmatrix} \cosh 2r & 0 & \sinh 2r & 0 \\ 0 & \cosh 2r & 0 & -\sinh 2r \\ \sinh 2r & 0 & \cosh 2r & 0 \\ 0 & -\sinh 2r & 0 & \cosh 2r \end{pmatrix} \tag{114}$$

3 State Discrimination Using Gaussian States

3.1 Introduction

With the basic formalism and notation for continuous variable states established, we now address the problem of discriminating between such states, focusing particularly on

Gaussian states.

A continuous variable state is classified as Gaussian if its Wigner function – or equivalently, its characteristic function – takes a Gaussian form. Mathematically, this is expressed as:

$$W[\hat{\rho}](\vec{R}) = \frac{1}{(2\pi)^n\sqrt{\det\sigma}} \exp\left(-\frac{1}{2}(\vec{R}-\bar{R})^T\sigma^{-1}(\vec{R}-\bar{R})\right) \tag{115}$$

where,

- $W[\hat{\rho}](\vec{R})$: The Wigner function tells us the 'probability-like' value at point $\vec{R}$ in phase space.
- $\vec{R}$: This is a vector containing the positions and momenta of all n modes. For example, for one mode, $\vec{R} = (q, p)$.
- $\bar{R}$: The average (mean) value of $\vec{R}$. This is the center of the bell curve.
- σ: The covariance matrix. It tells us how spread out the state is in different directions.
- $\det\sigma$: The determinant of the covariance matrix. It measures the overall volume of the distribution.
- n: The number of modes (like the number of particles or light beams).

Alternatively,

$$\chi[\hat{\rho}](\vec{\Lambda}) = \exp\left(-\frac{1}{2}\vec{\Lambda}^T\sigma\vec{\Lambda} + i\vec{\Lambda}^T\bar{R}\right) \tag{116}$$

where,

- $\chi[\hat{\rho}](\vec{\Lambda})$: The characteristic function. It is like the 'moment-generating' function in statistics.
- $\vec{\Lambda}$: A vector in phase space that is conjugate to $\vec{R}$ (like frequency is conjugate to time).
- σ: The same covariance matrix from Equation (3.1).
- $\bar{R}$: The same mean vector from Equation (3.1).

Thus, a Gaussian state $\hat{\rho}$ is completely characterized by its covariance matrix σ and its mean vector $\bar{R}$. The covariance matrix gains further physical significance through Williamson's Theorem, which decomposes it into a symplectic matrix and a diagonal

matrix of symplectic eigenvalues. Importantly, all entanglement properties of a Gaussian state are contained within σ. This is because local operations do not affect entanglement, so the mean vector $\bar{R}$ can be arbitrarily adjusted through phase space translations without altering the entanglement of $\hat{\rho}$. Conversely, generating entanglement necessarily requires a nonlocal operation on $\hat{\rho}$.

3.2 Normal form decomposition

A useful theorem for expressing the covariance matrix of a Gaussian state is Williamson's Theorem, stated as follows:

Theorem 3.1 (Williamson's Theorem)**.** *Given a real $2n \times 2n$ symmetric and positive matrix σ, there exists a real and symplectic $2n \times 2n$ matrix S and a real $n \times n$ diagonal and positive matrix W such that*

$$\sigma = S^T W S \tag{117}$$

where

$$W = \bigoplus_{k=1}^{n} d_k \mathbb{I}_2 \tag{118}$$

and

$$\mathbb{I}_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \tag{119}$$

Proof. The proof proceeds in several steps. We will construct the symplectic matrix S that diagonalizes σ in the symplectic sense.

Since σ is real, symmetric, and positive-definite, it can be diagonalized by an orthogonal matrix $O \in O(2n)$:

$$\sigma = O^T D O, \tag{120}$$

where $D = \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_{2n})$ with $\lambda_i > 0$. However, this is an orthogonal diagonalization, not a symplectic one. Our goal is to find a symplectic matrix S that brings σ to the block-diagonal form W.

Define the matrix

$$M = i\Omega\sigma, \tag{121}$$

where Ω is the symplectic form:

$$\Omega = \bigoplus_{k=1}^{n} \omega, \qquad \omega = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \tag{122}$$

Consider the matrix $M = i\Omega\sigma$. Its transpose is:

$$M^T = (i\Omega\sigma)^T = i\sigma^T\Omega^T = i\sigma(-\Omega) = -i\sigma\Omega. \tag{123}$$

Now note that:

$$\Omega^{-1}M\Omega = \Omega^{-1}(i\Omega\sigma)\Omega = i\sigma\Omega. \tag{124}$$

Since $\sigma\Omega$ is similar to $-\Omega\sigma$ (because $\Omega^T = -\Omega$), the eigenvalues of M come in $\pm$ pairs. With the factor i, the eigenvalues are of the form $\pm id_k$ where $d_k > 0$.

Let the eigenvalues of M be $\pm id_1, \pm id_2, \ldots, \pm id_n$ with $d_k > 0$. For each eigenvalue id_k, there exists a complex eigenvector u_k such that:

$$i\Omega\sigma u_k = id_k u_k \quad \Rightarrow \quad \Omega\sigma u_k = d_k u_k. \tag{125}$$

Similarly, for eigenvalue $-id_k$, we have an eigenvector v_k satisfying $\Omega\sigma v_k = -d_k v_k$.

Using the fact that σ is symmetric and Ω is skew-symmetric, one can show that v_k is proportional to the complex conjugate of u_k. We choose them such that:

$$v_k = i\sigma u_k. \tag{126}$$

This choice ensures the correct normalization.

We normalize the eigenvectors such that:

$$u_k^\dagger \sigma u_k = 1. \tag{127}$$

Then, using properties of Ω and σ, we also get:

$$u_k^\dagger \Omega u_k = i. \tag{128}$$

Define the real vectors x_k and y_k for $k = 1, \ldots, n$ as:

$$x_k = \sqrt{2}\ \mathrm{Re}(u_k), \qquad y_k = \sqrt{2}\ \mathrm{Im}(u_k). \tag{129}$$

One can verify that these vectors satisfy:

$$x_k^T \sigma x_k = y_k^T \sigma y_k = d_k, \tag{130}$$

$$x_k^T \sigma y_k = 0, \tag{131}$$

and

$$x_k^T \Omega x_k = y_k^T \Omega y_k = 0, \tag{132}$$

$$x_k^T \Omega y_k = 1. \tag{133}$$

Let S be the $2n \times 2n$ matrix whose columns are, in order:

$$S = \begin{pmatrix} x_1 & y_1 & x_2 & y_2 & \cdots & x_n & y_n \end{pmatrix}. \tag{134}$$

From the properties above:

- Symplectic condition: $S^T \Omega S = \Omega$ (since $x_k^T \Omega y_k = 1$ and all other pairings are zero).
- Diagonalization of σ: $S^T \sigma S = \bigoplus_{k=1}^n d_k \mathbb{I}_2 = W$.

Thus, we have constructed a symplectic matrix S such that:

$$S^T \sigma S = W \quad \Rightarrow \quad \sigma = (S^{-1})^T W S^{-1}. \tag{135}$$

Since S is symplectic, S^{-1} is also symplectic. Renaming S^{-1} as S (or absorbing the inverse into the definition), we obtain:

$$\sigma = S^T W S \tag{136}$$

with $S \in Sp(2n, \mathbb{R})$ and W diagonal as required. This completes the proof. □

The elements d_k of the matrix W are called the symplectic eigenvalues and can be calculated from the spectrum of $i\Omega\sigma$, while the matrix S is the symplectic matrix that performs the symplectic decomposition. Finding the symplectic eigenvalues of a covariance matrix is usually straightforward, but determining the symplectic matrix that performs the decomposition is often difficult. Performing the symplectic decomposition of Gaussian states is a key part of the process of determining the distinguishability of those states. For example, it turns out that in order to calculate the quantum Chernoff bound for Gaussian states – an asymptotic distinguishability measure for states that will be used extensively – we must find the symplectic decomposition of the states involved.

The phenomenological statement implied by equation (3.3) is that every Gaussian state $\hat{\rho}_G$ can be obtained from a thermal state $\hat{\rho}_{\text{th}}$ (described by a diagonal covariance matrix as in equation (2.96)) by performing the unitary transformation U_S associated with the symplectic matrix S that performs the symplectic decomposition:

$$\hat{\rho}_G = U_S \hat{\rho}_{\text{th}} U_S^\dagger \tag{137}$$

where $\hat{\rho}_{\text{th}} = \hat{\rho}_{\text{th},1} \otimes \hat{\rho}_{\text{th},2} \otimes \cdots \otimes \hat{\rho}_{\text{th},n}$ is a product of thermal states $\hat{\rho}_{\text{th},k}$ for $k = 1, \ldots, n$. Now, we derive the above equation.

From Williamson's Theorem, we have:

$$\sigma = S^T W S \tag{3.3}$$

where:

- σ is the covariance matrix of an arbitrary Gaussian state $\hat{\rho}_G$
- S is a symplectic matrix ($S \in Sp(2n, \mathbb{R})$)
- W is a diagonal matrix of the form:

$$W = \bigoplus_{k=1}^{n} d_k \mathbb{I}_2, \qquad \mathbb{I}_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \tag{138}$$

For a single-mode thermal state $\hat{\rho}_{\text{th},k}$ with average photon number N_k, the covariance matrix is (from equation 2.96):

$$\sigma_{\text{th},k} = \frac{1}{4} \begin{pmatrix} 2N_k + 1 & 0 \\ 0 & 2N_k + 1 \end{pmatrix} = \left(N_k + \frac{1}{2} \right) \mathbb{I}_2 \tag{139}$$

Comparing this with the block form of W, we identify:

$$d_k = N_k + \frac{1}{2} \tag{140}$$

Thus, the matrix W can be written as:

$$W = \bigoplus_{k=1}^{n} \left(N_k + \frac{1}{2} \right) \mathbb{I}_2 \tag{141}$$

This is exactly the covariance matrix of a **product of thermal states**:

$$\hat{\rho}_{\text{th}} = \hat{\rho}_{\text{th},1} \otimes \hat{\rho}_{\text{th},2} \otimes \cdots \otimes \hat{\rho}_{\text{th},n} \tag{142}$$

whose covariance matrix is:

$$\sigma_{\text{th}} = W \tag{143}$$

Now, we want to find a unitary transformation U_S that maps the thermal state $\hat{\rho}_{\text{th}}$ to the target Gaussian state $\hat{\rho}_G$.

In phase space, a unitary transformation generated by a quadratic Hamiltonian corresponds to a symplectic transformation on the covariance matrix (see Appendix C of

the thesis):

$$\sigma \to S\sigma S^T \tag{144}$$

More precisely, if $\hat{\rho}' = U_S \hat{\rho} U_S^\dagger$, then the covariance matrix transforms as:

$$\sigma' = S\sigma S^T \tag{145}$$

where S is the symplectic matrix associated with the unitary U_S.

Let us apply the symplectic transformation S to the thermal state $\hat{\rho}_{\text{th}}$:

$$\hat{\rho}_G = U_S \hat{\rho}_{\text{th}} U_S^\dagger \tag{146}$$

The covariance matrix of $\hat{\rho}_G$ becomes:

$$\sigma_G = S\sigma_{\text{th}} S^T \tag{147}$$

But $\sigma_{\text{th}} = W$, so:

$$\sigma_G = SWS^T \tag{148}$$

Williamson's Theorem states:

$$\sigma = S^T W S \tag{149}$$

Notice the difference: Williamson's theorem has $S^T W S$, while we have SWS^T. However, since S is symplectic, its inverse is given by:

$$S^{-1} = \Omega S^T \Omega^{-1} \tag{150}$$

More simply, if S is symplectic, then S^T is also symplectic. Therefore, we can rename S^T as a new symplectic matrix S'. This means:

$$SWS^T = (S^T)^T W (S^T) = S'^T W S' \tag{151}$$

Thus, both forms are equivalent. Williamson's theorem guarantees the existence of some symplectic matrix such that $\sigma = S^T W S$. Therefore, there exists a unitary transformation U_S such that:

$$\hat{\rho}_G = U_S \hat{\rho}_{\text{th}} U_S^\dagger \tag{152}$$

Thus, we arrive at the equation:

$$\hat{\rho}_G = U_S \hat{\rho}_{\text{th}} U_S^\dagger \tag{153}$$

where:

- $\hat{\rho}_G$ is any Gaussian state
- $\hat{\rho}_{\text{th}} = \hat{\rho}_{\text{th},1} \otimes \cdots \otimes \hat{\rho}_{\text{th},n}$ is a product of thermal states
- U_S is the unitary transformation associated with the symplectic matrix S

3.3 Distinguisability Of States

Although, the problem of distinguisability is defined for any number of states, let us start with the problem of distinguisability of two states, $\hat{\rho}_0$ and $\hat{\rho}_1$, with their corresponding equiprobable binary hypothesis H_0 and H_1 respectively. However, so far, we have only discussed more of the formalism, but nothing that could be used quantify the distinction between two states. In particular, we want something that can quantify some *distance* between the two states or some kind of method to quantify some *error* for distinction between two states given the measurement. For the notion of *distance*, we use *fidelity of states* and as discused in chapter 1, the *Helstrom's Bound* can be viewed as a measure of the error which gives the minimum probability of error and the corresponding POVM that gives the minimum measurement errors.

3.3.1 Helstrom's Optimal Discrimination

As discussed in chapter 1, Helstrom's Bounds corresponds to the generic optimization problem to find the POVM that gives the best measurement with minimum error probability. Given the two states, we consider a dichotomous POVM $\{\hat{E}_0, \hat{E}_1\}$, where $\hat{E}_0$ corresponds to the hypothesis $\hat{H}_0$ and $\hat{E}_1$ corresponds to the hypothesis $\hat{E}_1$ respectively. The error probability can simply be obtained from the hypothesis testing for H_0 and H_1 as

$$P_E = \frac{1}{2}(P(H_0|H_1) + P(H_1|H_0)) \tag{154}$$

where,
$P(H_0|H_1)$ is the probability of concluding that H_0 is true, when H_1 is and
$P(H_1|H_0)$ is the probability of concluding that H_1 is true, when H_0 is true and
the factor of $\frac{1}{2}$ arises from the equiprobable assumptions.

Also, note that the expectation of an operator $\hat{O}$ given the system is in state $\hat{\rho}$ is given by $\left\langle \hat{O} \right\rangle = tr(r\hat{h}o\hat{O})$. So, if the system is in $\hat{\rho}_j$ and we make a measurement with the operator with $\hat{E}_i$, then the probability of error in the measurement is given by

$$P(H_i|H_j) = tr(\hat{\rho}_j \hat{E}_i) \tag{155}$$

Thus, using eq 155 in eq 154, we get

$$P_E = \frac{1}{2}(tr(\hat{\rho}_1\hat{E}_0) + tr(\hat{\rho}_0\hat{E}_1)) \tag{156}$$

Also, since we defined our POVM to be the set $\hat{E}_0, \hat{E}_1$ and POVM are complete, so $\hat{E}_0 + \hat{E}_1 = I \implies \hat{E}_0 = I - \hat{E}_1$. So, eq 156 will be,

$$\begin{aligned} P_E &= \frac{1}{2}(tr(\hat{\rho}_1(I - \hat{E}_1)) + tr(\hat{\rho}_0\hat{E}_1)) \\ &= \frac{1}{2}(tr(\hat{\rho}_1) - tr(\hat{\rho}_1\hat{E}_1) + tr(\hat{\rho}_0\hat{E}_1)) \\ &= \frac{1}{2}(1 - tr(\hat{\rho}_1\hat{E}_1) + tr(\hat{\rho}_0\hat{E}_1)), \text{since } tr(\hat{\rho}_i) = 1 \\ &= \frac{1}{2}(1 - tr((\hat{\rho}_1 - \hat{\rho}_0)\hat{E}_1)) \end{aligned} \tag{157}$$

Note, all we aim to is find the correct POVM set so that the error probability is minimum. For this case, as $\hat{E}_0 + \hat{E}_1 = I$, so, we can express $\hat{E}_0$ in terms of $\hat{E}_1$. Thus, we have to minimize P_E over the measurement operator $\hat{E}_1$. Define $\hat{\gamma} = \hat{\rho}_1 - \hat{\rho}_0$. Clearly, $tr(\hat{\gamma}) = tr(\hat{\rho}_1) - tr(\hat{\rho}_0) = 1 - 1 = 0$. Since, both $\hat{\rho}_0$ and $\hat{\rho}_1$ are both positive and semi definite, so the eigenvalues of $\hat{\gamma}$ can be both positive or negative. Since both $\hat{\rho}_0$ and $\hat{\rho}_1$ are hermitian, so $\hat{\gamma}$ is also hermitian, hence it has the spectral decomposition,

$$\hat{\gamma} = \sum_k \mu_k |k\rangle \langle k| \tag{158}$$

where μ_k are the real eigenvalues. Now expanding $\hat{E}_1$ in the same basis

$$\hat{E}_1 = \sum_{k,l} e_{kl} |k\rangle \langle l| \tag{159}$$

Hence,

$$\begin{aligned} tr(\hat{E}_1\hat{\gamma}) &= tr\left(\sum_{k,l} e_{kl} |k\rangle \langle l| \cdot \sum_m \mu_m |m\rangle \langle m|\right) \\ &= \sum_{klm} e_{kl}\mu_m \langle l|m\rangle \langle m|k\rangle \\ &= \sum_{klm} e_{kl}\mu_m \delta_{l,m}\delta_{m,k} \\ &= \sum_k e_{kk}\mu_k \end{aligned} \tag{160}$$

Since POVMs are positive valued, so e_{kk} is positive. Now, μ_k can be positive or negative, but since we are optimizing e_{kk}, if μ_k is negative, it will lead to negative propagation

of the errors, thus for such cases e_{kk} must be 0. Hence the net contribution will come only from the positive e_{kk} values. Thus the optimal $\hat{E}_1$ is one that projects onto positive part of $\hat{\gamma}$, $\hat{\gamma}_+$ (say) which projects to the subspace by the eigenstates of $\hat{\gamma}$ with positive eigenvalues. Thus, we have $tr(\hat{\gamma}\hat{E}_1) = tr(\hat{\gamma}_+) = \frac{1}{2}||\hat{\gamma}||_1$ where $||\hat{\gamma}||_1 = tr(|\hat{\gamma}|) = tr(\sqrt{\gamma^\dagger\gamma})$. Thus, using this, eq 157 can be written as

$$P_E^{min} = \frac{1}{2}(1 - ||\hat{\rho}_1 - \hat{\rho}_0||_1) \tag{161}$$

However, the calculation of the minimum probability of error using the above hypothesis is only for a single particle system with two states. But, the same analysis and calculations for an N particle system would be very difficult. This is because as the number of particles in the system increases the Hilbert Space grows exponentially over N as the N copy density matrix resides in the tensor product state of dimension d^N, where d is the dimension (degrees of freedom) for a single particle state, which may not simplify for larger N values. Moreover, opimization of such d^N matrix set (POVM) would also be analytically as well as numerically very difficult. However, one can calculate the *Quantum Chernoff Bounds*, which gives the asymptotic dependency of N copys of the system on the probability of error. Then the quantum hypothesis test becomes

$$\begin{aligned} H_0 &: \hat{\rho}_0 \otimes ... \otimes \hat{\rho}_0 \\ H_1 &: \hat{\rho}_1 \otimes ... \otimes \hat{\rho}_1 \end{aligned} \tag{162}$$

Thus for this case, the minimum probability of error is given by

$$P_{E,N}^{min} = \frac{1}{2}(1 - ||\hat{\gamma}_N||_1) \tag{163}$$

where $\hat{\gamma}_N = \hat{\rho}_1 \otimes ... \otimes \hat{\rho}_1 - \hat{\rho}_0 \otimes ... \otimes \hat{\rho}_0$

3.3.2 Fidelity

Fidelity between two states could be used as a notion of measure of the *distance* between two states. For two states, $\hat{\rho}_1$ and $\hat{\rho}_2$, fidelity is defined as,

$$F(\hat{\rho}_1, \hat{\rho}_2) = \left(tr\left(\sqrt{\sqrt{\hat{\rho}_1}\hat{\rho}_2\sqrt{\hat{\rho}_1}}\right)\right)^2 \tag{164}$$

The properties of the above fidelity equation to be used as a viable measure of distance between two states are:

- F($\hat{\rho}_1$, $\hat{\rho}_2$) ≤ 1
- F($\hat{\rho}_1$, $\hat{\rho}_2$) = F($\hat{\rho}_2$, $\hat{\rho}_1$)

- If $\hat{\rho}_1$ is a pure state, $\hat{\rho}_1 = |\psi_1\rangle\langle\psi_1|$, then $\mathrm{F}(\hat{\rho}_1, \hat{\rho}_2) = \langle\psi_1|\hat{\rho}_2|\psi_1\rangle$
- $\mathrm{F}(\hat{\rho}_1, \hat{\rho}_2)$ is invariant under unitary transformations on the state space.
- $\mathrm{F}(\hat{\rho}_1 \otimes \hat{\sigma}_1, \hat{\rho}_2 \otimes \hat{\sigma}_2) = \mathrm{F}(\hat{\rho}_1, \hat{\rho}_2)\ \mathrm{F}(\hat{\sigma}_1, \hat{\sigma}_2)$

3.3.3 Quantum Chernoff Bounds

So far, we have discussed how the error is quantified and methods of quantifying the error. But, how is the actual minimization of the errror probability occurs. Chernoff solved this problem in the asymptotic regime and gave the asymptotic dependence on the probability of error if we have N copies of the system, i.e. a N trail hypothesis testing.[2] He showed that the probability of error decreases exponentially with the number of tests performed N, i.e. $P_E \approx exp(-\kappa N)$. The optimal exponent κ arising in the asymptotic limit is called Chernoff Bounds[2]. But why is Chernoff Bounds even considered... This may go back to 'One of the virtues of the Chernoff bound is that it yields a very natural distance measure between probability distributions; it is essentially the unique distance measure in the ubiquitous situation of independent and identically distributed (i.i.d.) random variables.'[2]

The quantum chernoff bounds serves as the upper limit the the minimum probability of error 163, i.e.

$$P_{N,E}^{min} \leq P_{QC}^{N} = \frac{1}{2} exp(-\kappa N) \tag{165}$$

where $\kappa = -ln\left(\min_{0\leq s\leq 1} tr(\hat{\rho}_0^s \hat{\rho}_1^{(1-s)})\right)$

Before diving deep into the quantum chernoff bounds, let us discuss a bit of the classical chernoff bounds. Consider a random variable X and $a \in \mathbb{R}$. Then, [3]

$$\begin{aligned} P(X \geq a) &\leq min_{s>0} e^{-sa} M_X(s) \\ P(X \leq a) &\leq min_{s<0} e^{-sa} M_X(s) \end{aligned} \tag{166}$$

For the optimal decision, we define the log likelihood,

$$L(x) = log \frac{p_1(x)}{p_0(x)}$$

For N trial observations

$$X = \sum_{i=1}^{N} L(x_i) = \sum_{i=1}^{N} log \frac{p_1(x_i)}{p_0(x_i)}$$

where we declare H_1 is true if X ¿ 0 else we declare H_0 is true. Now we refer to the eq 154, which using above under hypothesis H_1

$$P(H_0|H_1) = P(X \leq 0|H_1) \leq e^{-s\cdot 0} M_X(s) = M_X(s)$$

where $s < 0$ The moment generating function $M_X(s)$ is defined as $\langle e^{sX} \rangle$, Hence

$$\begin{aligned} M_X(s) &= \langle e^{sX} \rangle = \langle e^{s\sum_{i=1}^{N} log\frac{p_1(x_i)}{p_0(x_i)}} \rangle \\ M_X(s) &= \langle e^{\sum_{i=1}^{N} log\frac{p_1(x_i)}{p_0(x_i)}^s} \rangle \\ M_X(s) &= \langle \prod_{i=1}^{N} e^{log\frac{p_1(x_i)}{p_0(x_i)}^s} \rangle \\ M_X(s) &= \prod_{i=1}^{N} \langle \frac{p_1(x_i)}{p_0(x_i)}^s \rangle \end{aligned} \tag{167}$$

Since, each factor is identical and $\langle f \rangle = \sum_{x_i} p(x)f(x)$ so,

$$M_X(s) = \left(\sum_x p_1(x) \left(\frac{p_1(x)}{p_0(x)} \right)^s \right)^N = \left(\sum_x p_1(x)^{1+s} p_0(x)^{-s} \right)^N \tag{168}$$

Taking $s \to -s$, i.e. $s > 0$

$$M_X(s) = \left(\sum_x p_1(x)^{1-s} p_0(x)^s \right)^N \tag{169}$$

$$P(H_0|H_1) \leq \left(\sum_x p_1(x)^{1-s} p_0(x)^s \right)^N \tag{170}$$

Similarly for $P(H_1|H_0)$, we get

$$P(H_1|H_0) \leq \left(\sum_x p_1(x)^{1-s} p_0(x)^s \right)^N \tag{171}$$

where $s > 0$ Combining eq 170 and eq 171 with eq 154

$$P_E^N \leq \left(\sum_x p_1(x)^{1-s} p_0(x)^s \right)^N \tag{172}$$

Notice this takes the same form as the quantum Chernoff Bounds upto a factor of $\frac{1}{2}$, where instead of the probability desity functions $p_0(x)$ and $p_1(x)$, we have the density matrices ρ_0 and ρ_1.

One of the notable observation is that the Chernoff Bounds are exponentially tight. As the number of trails increases i.e. $N \to \infty$, $ln(P_{N,E}^{min})/N \to -\kappa$

Also, defining $Q_s = tr(\hat{\rho}_0^s \hat{\rho}_1^{1-s})$ and $Q = \min_{0 \leq s \leq 1} Q_s$, the quantum Chernoff Bounds can be written as

$$P_{QC}^N = \frac{1}{2} Q^N \tag{173}$$

The aim to find the minimum error would require the minimization of Q_s over all valures of $s \in [0,1]$. Setting s = $\frac{1}{2}$, we will arrive at the *Quantum Bhattacharyya Bound*, however this bound is not assymptotically tight as the Quantum Chernoff Bound.

$$P_B^N = \frac{1}{2}\left(tr(\sqrt{(}\hat{\rho}_0\hat{\rho}_1))\right)^N \tag{174}$$

If the two states are indistinguisable, i.e. $\hat{\rho}_0 = \hat{\rho}_1$, then $P_{QC}^N \approx P_B^N$ and

$$\frac{1-\sqrt{1-F(\hat{\rho}_0,\hat{\rho}_1)}}{2} \leq P^{(1)} \leq P_{QC}^{(1)} \leq \frac{\sqrt{F(\hat{\rho}_0,\hat{\rho}_1)}}{2}$$

Also, it is noted that Q_s is a convex function over s and joint-concave over $\hat{\rho}_0$ and $]r\hat{h}o_1$. Q_s can be written as

$$Q_s = tr(\hat{\rho}_0^s]r\hat{h}o_1^{1-s}) = \sum_k e_k^{(0)^s} e_k^{(1)^{1-s}}$$

where $e_k^{(0)}$ and $e_k^{(1)} \geq 0$ Thus each element of Q_s is $f_k(s) = a_k^s b_k^{1-s} = e^{slna_k} e^{(1-s)lnb_k}$, taking the 2nd derivative of f_k w.r.t. s, we see $f_k^{''}(s) = (lna_k - lnb_k)^s \cdot f_k(s) \geq 0$. Hence Q_s is convex over s. Also using Lieb's Theorem [4]

Theorem 3.2. *Lieb's Theorem: Let K be a fixed matrix in $\mathbb{C}^{n\times m}$. Then for any $t \in [0,1]$, the map*

$$(A,B) \to tr[K^* A^{1-t} K B^t]$$

is jointly concave in (A, B) where A and B are respectively $n \times n$ and $m \times m$ Hermitian positive definite matrices.

Taking $A = \hat{\rho}_0$ and $B = \hat{\rho}_1$, and $K = \mathbb{I}$ and $t = s$, then Q_s is concave over $\hat{\rho}_0$ and $\hat{\rho}_1$.

3.4 Fidelity for Gaussian States

3.4.1 Single Mode Gaussian States

For two single mode gaussian thermal states with mean 0 and covariance matrices σ_1 and σ_2 corresponding to states $\hat{\rho}_1$ and $\hat{\rho}_2$ is given by

$$F_0(\hat{\rho}_1,\hat{\rho}_2) = \frac{2}{\sqrt{det(\sigma_1+\sigma_2)}(\sqrt{det(\theta)} - \sqrt{det(\theta)-1})} \tag{175}$$

where,
$det(\theta) = 1 + \frac{P}{det(\sigma_1+\sigma_2)}$
$P = (det(\sigma_1)-1)(det(\sigma_2)-1)$
It was also derived in the paper, the fidelity for a two single mode displaced squeezed thermal states. By the Williamson's Theorem, it could be seen that any arbitrary

single mode gaussian state can be expressed as a displaced squeezed thermal state, i.e. $\hat{\rho}_G = \hat{U}_S \hat{\rho}_{th} \hat{U}_S^\dagger$. Thus, the fidelity for any single mode gaussian state,

$$F(\hat{\rho}_1, \hat{\rho}_2) = F_0(\hat{\rho}_1, \hat{\rho}_2) exp\left(-\frac{1}{2}(\overline{R_1} - \overline{R_2})^T \kappa (\overline{R_1} - \overline{R_2})\right) \tag{176}$$

where $F_0(\hat{\rho}_1, \hat{\rho}_2)$ is defined as eq 175 and $\kappa = \mathbb{I} - (\Phi(\sigma_1) + \sigma_2)^{-1}(\sigma_2 + i\Omega)$

3.4.2 Multimode Gaussian States

For comparing two multi-mode gaussian states, the fidelity is defined as [**multimode**]

$$F(\hat{\rho}_1, \hat{\rho}_2) = \sqrt{Ldet(\Phi(\mathcal{O}))}$$

where,
$\Phi(A) = A(\mathbb{I} + \sqrt{\mathbb{I} + (JA)^{-1}})$
$L^{-1} = det(\Phi(\sigma_1))^{-1} det(\frac{\Phi(\sigma_1)+\sigma_2}{2}) det(\frac{\sigma_2+\Phi(\sigma_1)-U}{2})$
$U = (\sigma_1 - i\Omega)(\Phi(\sigma_1) + \sigma_2)^{-1}(\sigma_2 + i\Omega)$
$\mathcal{O} = \Phi(\sigma_1) - (\Phi(\sigma_1) - i\Omega)\left(\sigma_2 + \Phi(\sigma_1) - (\sigma_2 - i\Omega)(\Phi(\sigma_1) + \sigma_2)^{-1}(\sigma_2 + i\Omega)\right)^{-1}(\Phi(\sigma_1) + i\Omega)$

3.5 Quantum Chernoff Bounds for Gaussian States

Consider two n-mode Gaussian states, viz $\hat{\rho}_A$ and $\hat{\rho}_B$ with symplectic decomposition matrix S_A and S_B respectively, with symplectic eigenvalues $\{\alpha_k, k = 1, ..., n\}$ and $\{\beta_k, k = 1, ..., n\}$ respectively. The mean in the phase space are $\bar{x_A}$ and $\bar{x_B}$ respectively.

We know by Williamson's Theorem that every n-mode gaussian state ρ can be specified by a displacement vector $\bar{x} \in \mathcal{R}^{2n}$, a symplectic matrix S that preserves Ω and the symplectic eigenvalues $\nu_k, k = 1, ..., n$, where $\nu_k \geq 1$ and via the dicomposition

$$\rho = \hat{U}_{\bar{x}S} \rho_{th} \hat{U}_{\bar{x}S}^\dagger$$

where $\rho_{th} = \bigotimes_{k=1}^n \sigma\nu_k$ is the thermal state. Now, $\sigma(\nu_k) = \frac{2}{\nu_k+1} \sum_{j=0}^\infty \left(\frac{\nu_k-1}{\nu_k+1}\right)^j |j\rangle \langle j|$. The covariance matrix is $V = S\left[\bigoplus_{k=1}^n \nu_k I_k\right] S^\dagger$[**gaussian˙states**][eq23, eq32]

Using *Lemma 1* of [**gaussian˙states**],

$$\rho^p = tr(\rho^p)\rho(p)$$

where $tr(\rho^p) = \prod_{k=1}^n G_p(\nu_k) = \prod_{k=1}^n \frac{2^p}{(\nu_k+1)^p - (\nu_k-1)^p}$. [**gaussian˙states**][eq30]

Then,

$$Q_s = tr(\hat{\rho}_A^s \hat{\rho}_B^{1-s}) = \prod_{k=1}^n G_s \alpha_k G_{1-s}(\beta_k) tr(\hat{\rho}_A(s)\hat{\rho}_B(1-s)) \tag{177}$$

with covariance matrices $V_A(s)$ and $V_B(1-s)$.

Also, from chapter 2 we have

$$tr(\hat{O}_1, \hat{O}_2) = \int_{\mathbb{C}^n} \frac{d^{2n}\vec{\lambda}}{\pi^n} \chi[\hat{O}_1](\vec{\lambda})\chi[\hat{O}_2](-\vec{\lambda})$$

Using the characteristics function for the gaussian state $\chi(\vec{\lambda}) = exp(-\frac{1}{2}\vec{\lambda}^T\sigma\vec{\lambda} + i\vec{\lambda}^T\bar{R})$ we get,

$$tr(\hat{\rho}_A, \hat{\rho}_B) = \int_{\mathbb{C}^n} \frac{d^{2n}\vec{\lambda}}{\pi^n} exp\left(-\frac{1}{2}\vec{\lambda}^T(V_A + V_B)\vec{\lambda} + i(\bar{x}_A - \bar{x}_B)\vec{\lambda}\right) \tag{178}$$

This is a standard integral. Solving we get,

$$tr(\hat{\rho}_A, \hat{\rho}_B) \propto exp\left(-\frac{1}{2}(\vec{x}_A - \vec{x}_B)^T(V_A + V_B)^{-1}(\vec{x}_A - \vec{x}_B)\right)$$

$$tr(\hat{\rho}_A, \hat{\rho}_B) \propto exp\left(-\frac{1}{2}\vec{d}^T(V_A + V_B)^{-1}\vec{d}\right)$$

where $\vec{d} = \vec{x}_A - \vec{x}_B$ Hence,

$$Q_s = tr(\hat{\rho}_A^s\hat{\rho}_B^{1-s}) \propto \prod_{k=1}^{n} G_s\alpha_k G_{1-s}(\beta_k) exp\left(-\frac{1}{2}\vec{d}^T(V_A + V_B)^{-1}\vec{d}\right)$$

$$Q_s = tr(\hat{\rho}_A^s\hat{\rho}_B^{1-s}) = \bar{Q}_s exp\left(-\frac{1}{2}\vec{d}^T(V_A + V_B)^{-1}\vec{d}\right)$$

It can further be calculated to find the value of $\bar{Q}_s$ as

$$\bar{Q}_s = \frac{2^n \prod_{k=1}^{n} G_s(\alpha_k)G_{1-s}(\beta_k)}{\sqrt{\det\left(V_A(s) + V_B(1-s)\right)}}$$

where,

$$V_A(s) = S_A\left(\bigotimes_{k=1}^{n} \Lambda_s(\alpha_k)\mathbb{I}_k\right) S_A^T$$

$$V_B(1-s) = S_B\left(\bigotimes_{k=1}^{n} \Lambda_{1-s}(\beta_k)\mathbb{I}_k\right) S_B^T$$

$$G_p(x) = \frac{2^p}{(x+1)^p - (x-1)^p}$$

$$\Lambda_p(x) = \frac{(x+1)^p + (x-1)^p}{(x+1)^p - (x-1)^p}$$

The above implementation can be used numerically to calculate the quantum Chernoff Bounds. However, the analytical solution, for most cases would be difficult because the minimization over s, which is required would be difficult for most physical systems due to

the scale of the physical systems.

3.5.1 Numerical Analysis of Quantum Chernoff Bounds

Below is a numerical implimentation for the validation of the Helstrom Bounds over the limit of N.

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import minimize_scalar
from qutip import rand_dm, ket2dm, tensor
```

Listing 1: Includes

Defining the two random density matrices

$$\hat{\rho}_0 = \begin{pmatrix} 0.5276 & -0.4914 + 0.0466i \\ -0.4914 - 0.0466i & 0.4724 \end{pmatrix}$$

$$\hat{\rho}_1 = \begin{pmatrix} 0.4214 & 0.1045 + 0.2534i \\ 0.1045 - 0.2534i & 0.5786 \end{pmatrix}$$

```
rho0 = rand_dm(2, seed=809)
rho1 = rand_dm(2, seed=593)
print(rho0)
print(rho1)
```

Listing 2: Defining the two state with random density matrix

```
def mat_power(rho, t):
    vals, vecs = rho.eigenstates()
    vals = np.maximum(vals, 0)
    return sum(v**t * ket2dm(vec) for v, vec in zip(vals, vecs))
```

Listing 3: Defining a function to calculate the matrix power of rho

```
def Qs(r0, r1, s):
    return np.real((mat_power(r0, s) * mat_power(r1, 1-s)).tr())
```

Listing 4: Defining a function to return the Qs

```
# Minimize Qs wrt s within the bounds 0 < s < 1
res = minimize_scalar(lambda s: Qs(rho0, rho1, s), bounds=(0.01,0.99),
    method='bounded')
```

Listing 5: Gaussian state discrimination with QuTiP: Helstrom bound and Quantum Chernoff bound

```python
# Extract the results
Q = res.fun
s_opt = res.x
Q_half = Qs(rho0, rho1, 0.5)
xi = -np.log(Q)
print(f"s*={s_opt:.4f}  Q={Q:.6f}  Q_half={Q_half:.6f}  xi={xi:.6f}")
```

Listing 6: Extracting the results

```python
def helstrom_N(r0, r1, N):
    r0N   = tensor([r0]*N)
    r1N   = tensor([r1]*N)
    Gamma = r1N - r0N
    evals = Gamma.eigenenergies()
    del r0N, r1N, Gamma
    return 0.5*(1.0 - 0.5*np.sum(np.abs(evals)))
```

Listing 7: Defining a function to return the N trails accumulated helstrom bounds

```python
N_max  = 10
N_vals = np.arange(1, N_max+1)
print("Computing Helstrom...")
PE_exact = np.array([helstrom_N(rho0, rho1, N) for N in N_vals])
print("Done:", PE_exact)
```

Listing 8: Computing the helstrom bounds

```python
PE_QCB   = 0.5 * Q**N_vals
PE_bhat  = 0.5 * Q_half**N_vals
log_exact = np.log(np.maximum(PE_exact,1e-15)) / N_vals
log_QCB   = np.log(PE_QCB)  / N_vals
log_bhat  = np.log(PE_bhat) / N_vals
```

Listing 9: Calculating the probabilities of error

```python
theta_vals = np.linspace(0,1,40)
pe_h, pe_q, pe_b, pe_f = [],[],[],[]
for th in theta_vals:
    r1t = th*rho1 + (1-th)*rho0
    ev  = (r1t-rho0).eigenenergies()
    pe_h.append(0.5*(1-0.5*np.sum(np.abs(ev))))
    res2 = minimize_scalar(lambda s,r=r1t: Qs(rho0,r,s),bounds
   =(0.01,0.99),method='bounded')
    pe_q.append(0.5*res2.fun)
    pe_b.append(0.5*Qs(rho0,r1t,0.5))
    r0h = mat_power(rho0,0.5)
    F   = float(np.real(((r0h*r1t*r0h).sqrtm()).tr()))
    pe_f.append((1-np.clip(F,0,1))/2)
```

Listing 10: Finding the probability of error for different thetas over different bounds

```
plt.figure(figsize=(10, 7))
s_values = np.linspace(0.01, 0.99, 1000)
Q_values = [Qs(rho0, rho1, s_) for s_ in s_values]
plt.plot(s_values, Q_values)
plt.xlabel('s')
plt.ylabel(R'$Q_s$')
plt.savefig('../figures/Q_vs_s.png', dpi=300)
plt.show()


plt.figure(figsize=(10, 7))
plt.axhline(-xi, label='Assymptotic Limit')
plt.plot(N_vals, log_exact, label='Exact')
plt.plot(N_vals, log_QCB, label='Quantum Chernoff bound')
plt.plot(N_vals, log_bhat, label='Bhattacharyya bound')
plt.xlabel('N')
plt.ylabel(R'$\frac{ln(P_E)}{N}$')
plt.grid()
plt.legend()
plt.savefig('../figures/QCB.png', dpi=300)
plt.show()


plt.figure(figsize=(10, 7))
plt.plot(theta_vals,pe_h,'-',lw=2.5,label='Exact Helstrom')
plt.plot(theta_vals,pe_q,'--',lw=2,  label='Quantum Chernoff bound')
plt.plot(theta_vals,pe_b,':',lw=2,  label='Bhattacharyya bound')
plt.plot(theta_vals,pe_f,'-.',lw=2,  label='Fidelity lower bound')
plt.xlabel(R'$\theta$')
plt.ylabel('Error Probability')
plt.legend()
plt.savefig('../figures/P_E.png', dpi=300)
plt.show()
```

Listing 11: Plotting

3.5.2 Results

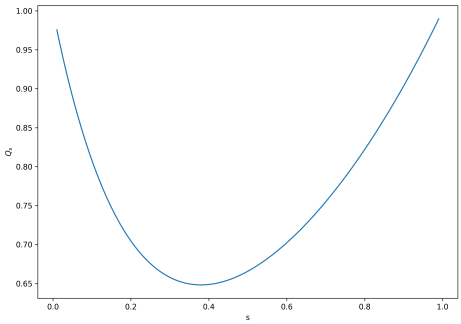

Figure 2: Convex nature of Qs

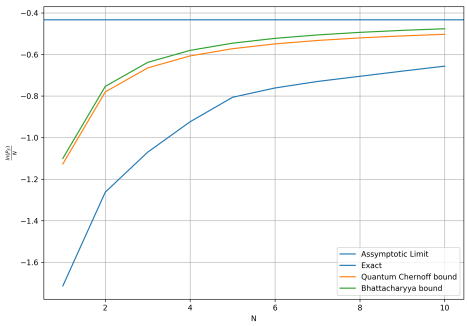

Figure 3: Different Bounds as N increases

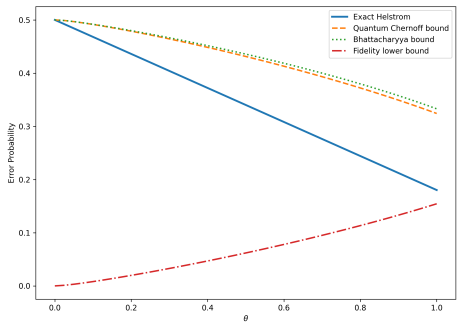

Figure 4: Error Probability within Bounds

References

[1] Si Hui Tan. "Quantum State Discrimination with Bosonic Channels and Gaussian States". In: *MIT Doctoral Thesis* (2010), pp. 0–70.

[2] R. Munoz-Tapia K.M.R. Audenaert; J. Calsamiglia and E. Bagan; Ll. Masanes; A. Acin; F. Verstraete. "Discriminating States: The Quantum Chernoff Bound". In: *Phys. Rev. Lett. 98, 160501 (2007)* (2007).

[3] Hossein Pishro-Nik. *Introduction to Probability, Statistics, and Random Processes: Chernoff Bounds*. URL: `https://www.probabilitycourse.com/chapter6/6_2_3_chernoff_bounds.php`.

[4] James Saunderson Hamza Fawzi. "Lieb's concavity theorem, matrix geometric means, and semidefinite optimization". In: *arXiv:1512.03401v3* (2020).